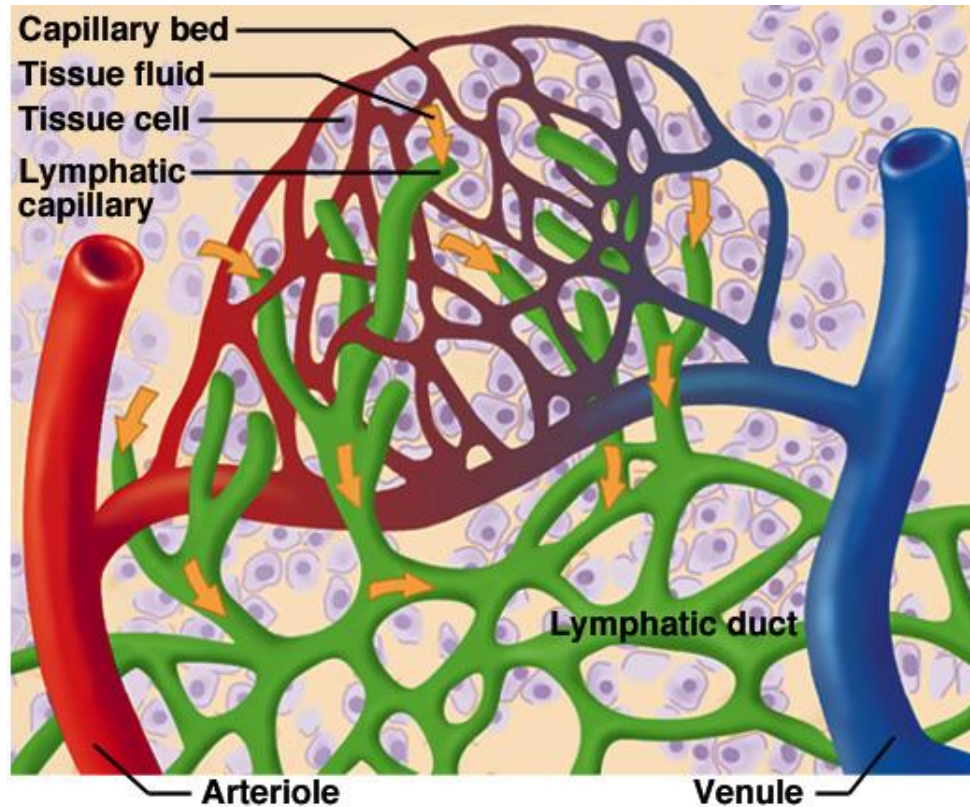
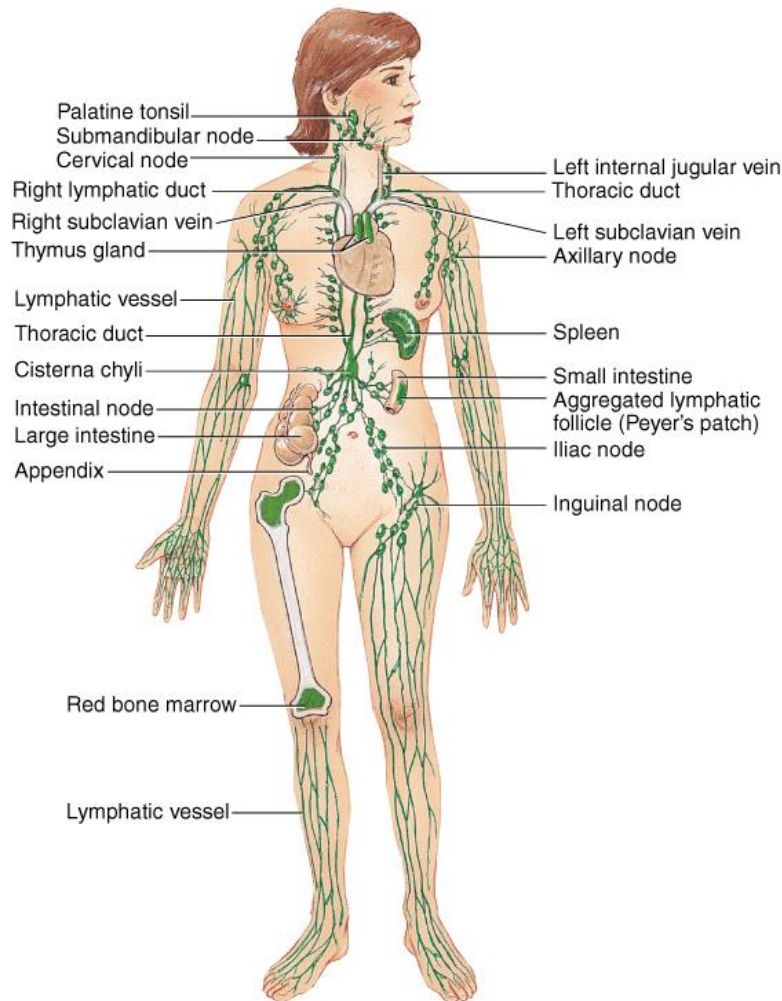


Lymphatic and Immune Systems



- Maintain fluid balance
- Protect body from infection and disease

Lymphatic System



- Organs, vessels and a fluid called lymph
 - similar to interstitial fluid
- Organs involved
 - red bone marrow
 - thymus
 - spleen
 - lymph nodes
 - diffuse lymphatic tissue
 - tonsils, adenoids & peyers patches

Functions of Lymphatic System

- Immunity
 - fluids from all capillary beds are filtered
 - immune cells stand ready to respond to foreign cells or chemicals encountered
- Lipid absorption
 - Lacteals in small intestine absorb dietary lipids
- Fluid recovery
 - absorbs plasma proteins and fluid (2 to 4 L/day) from tissues and returns it to the bloodstream
 - interference with lymphatic drainage leads to severe edema

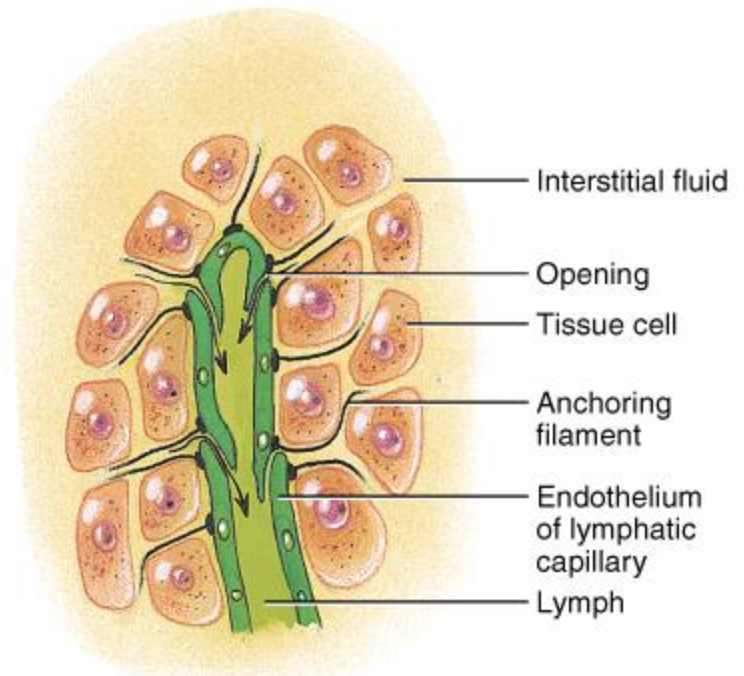


Lymph and Lymphatic Capillaries

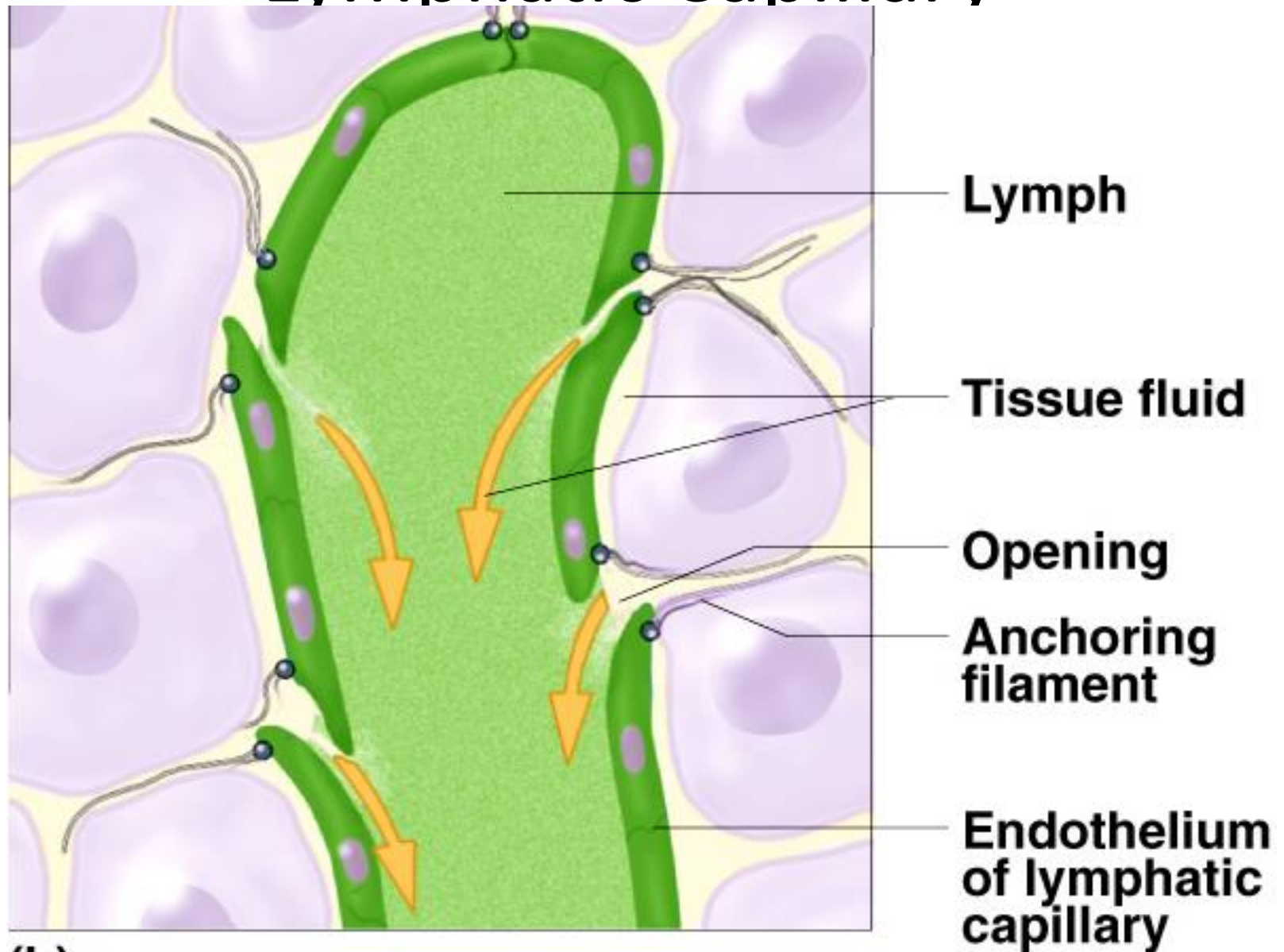
- Lymph
 - clear, colorless fluid, similar to plasma but much less protein
- Lymphatic capillaries
 - closed at one end
 - tethered to surrounding tissue by protein filaments
 - endothelial cells loosely overlapped
 - allow bacteria and cells entrance to lymphatic capillary
 - creates valve-like flaps that open when interstitial fluid pressure is high, and close when it is low

Lymphatic Capillaries

- Found throughout the body except in Avascular tissue (cartilage, epidermis & cornea)
- Structure is designed to let tissue fluid in but not out
 - anchoring filaments keep tube from collapsing under outside pressure
 - overlapping endothelial cells open when tissue pressure is high (one-way valve)
- In GI tract, known as lacteals -- contain chyle



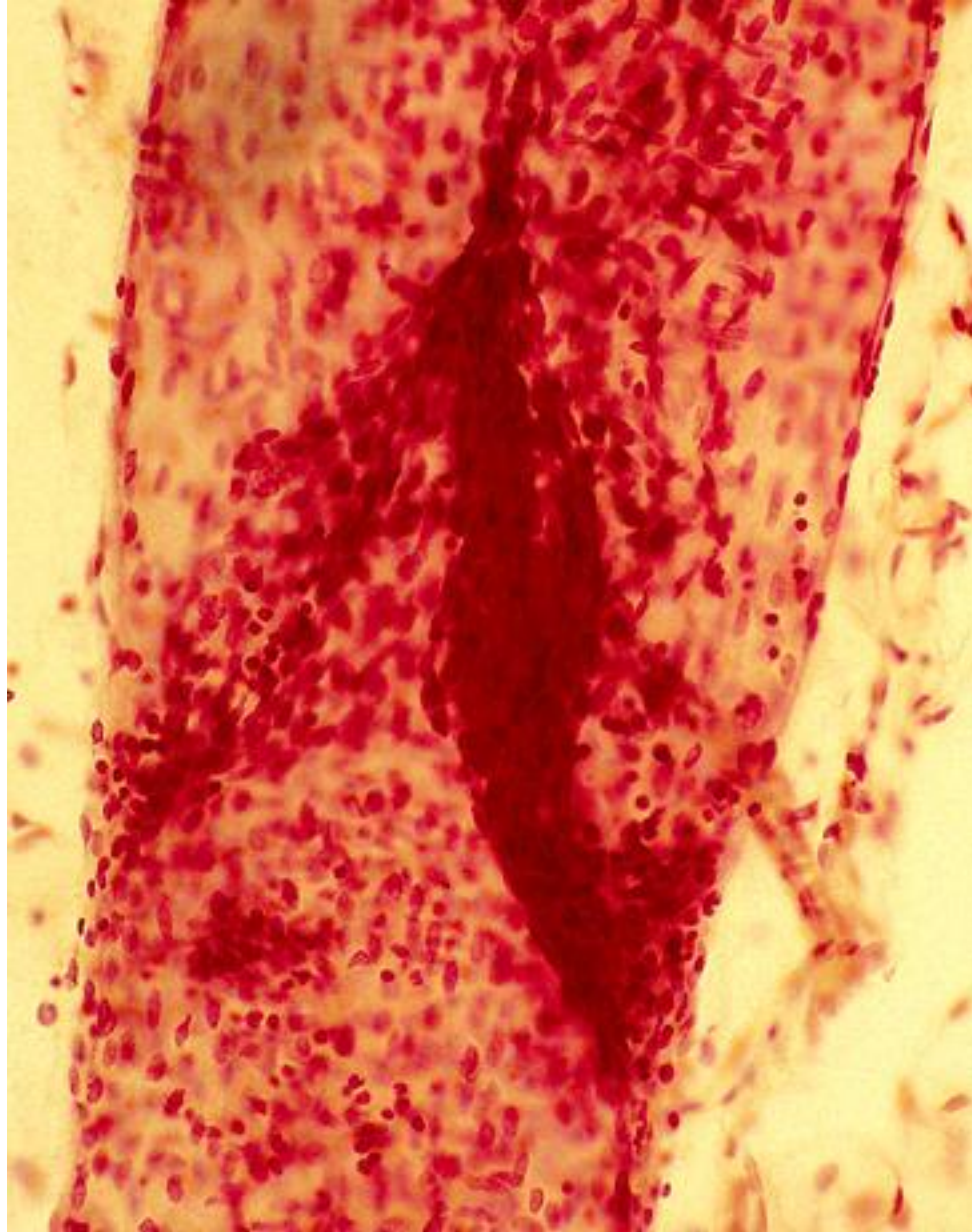
Lymphatic Capillary



Lymphatic Vessels

- Larger ones composed of 3 layers
 - tunica interna: endothelium and valves
 - tunica media: elastic fibers, smooth muscle
 - tunica externa: thin outer layer

Valve in a Lymphatic Vessel



Route of Lymph Flow

- Lymphatic capillaries
- Collecting vessels: course through many lymph nodes
- Lymphatic trunks: drain major portions of body
- Collecting ducts :
 - **right lymphatic duct** – receives lymph from R arm, R side of head and thorax; empties into R subclavian vein
 - **thoracic duct** - larger and longer, begins as a prominent sac in abdomen called the cisterna chyli, receives lymph from below diaphragm, left arm, left side of head, neck and thorax; empties into L subclavian vein

The Fluid Cycle

Lymphatic capillaries

Lymph node

Collecting duct

Lymphatic trunks

Lymph node

Collecting vessel

Lymphatic capillaries

Blood plasma

Tissue fluid

Lymph



Mechanisms of Lymph Flow

- Lymph flows at low pressure and speed
- Moved along by rhythmic contractions of lymphatic vessels-stretching of vessels stimulates contraction
- Flow aided by skeletal muscle pump
- Thoracic pump aids flow from abdominal to thoracic cavity
- Valves prevent backward flow
- Rapidly flowing bloodstream in subclavian veins, draws lymph into it
- Exercise significantly increases lymphatic return

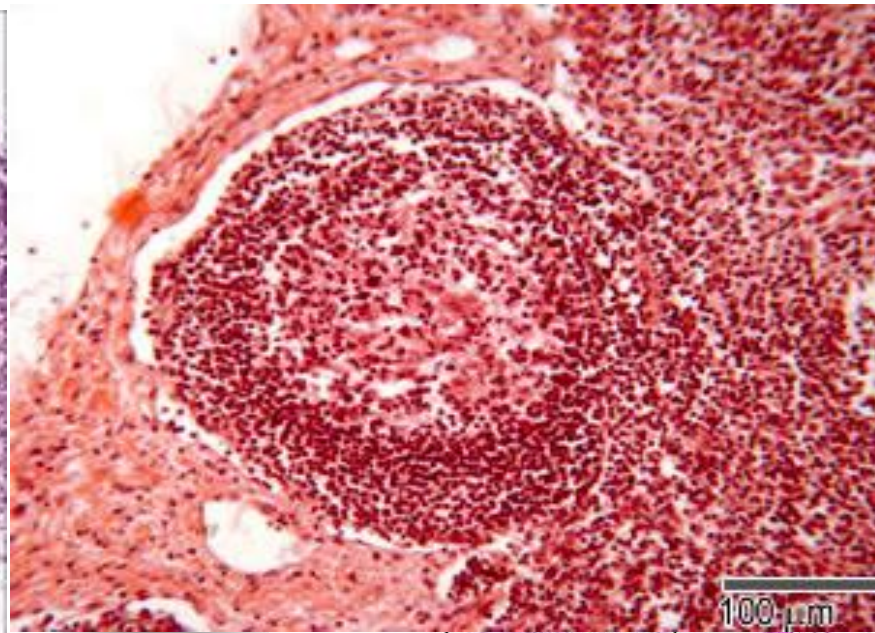
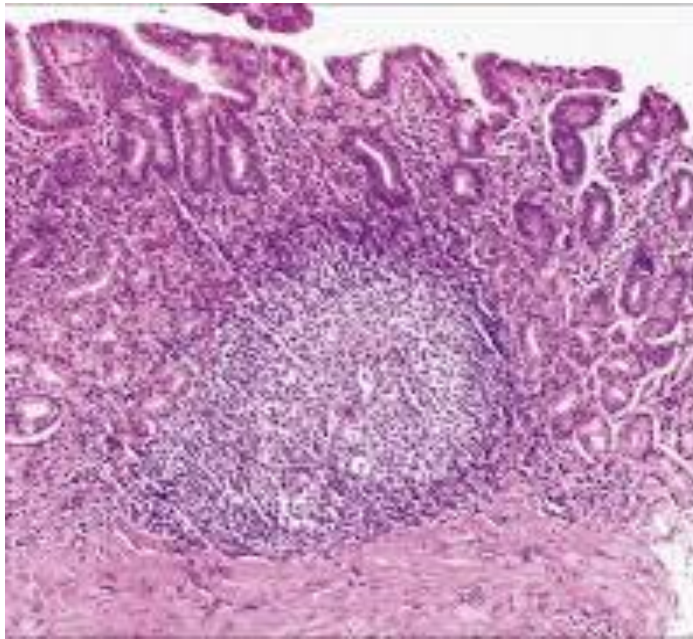
Lymphatic Cells

- T lymphocytes
 - Mature in thymus
- B lymphocytes
 - Activation causes proliferation and differentiation into plasma cells that produce antibodies
- Antigen Presenting Cells
 - Macrophages (from monocytes)
 - dendritic cells (in epidermis, mucous membranes and lymphatic organs)
 - reticular cells (also contribute to stroma of lymph organs)

Lymphatic Tissue

- Diffuse lymphatic tissue: lymphocytes in mucous membranes and CT of many organs
 - Mucosa-Associated Lymphatic Tissue: particularly prevalent in passages open to the exterior
- Lymphatic nodules: dense oval masses of lymphocytes, congregate in response to pathogens
 - Peyer patches: more permanent congregation, clusters found at junction of small to large intestine

Lymphatic tissues



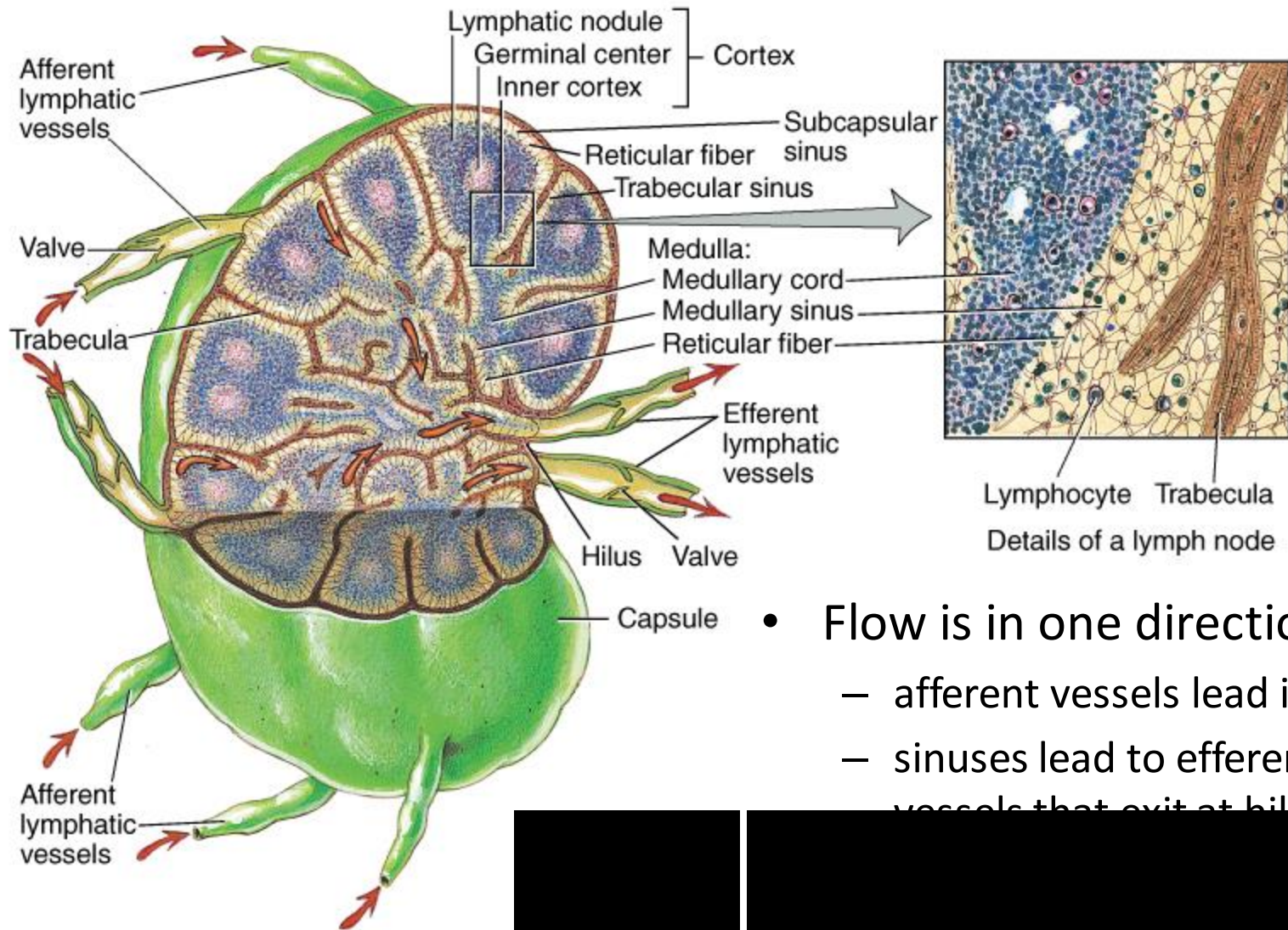
Lymphatic Organs

- At well defined anatomical sites, have CT capsules
- Lymph nodes
 - cervical, axillary and inguinal regions close to surface
 - thoracic, abdominal and pelvic groups deep in cavities
- Tonsils
 - guard entrance to pharynx
- Thymus
 - between sternum and aortic arch
- Spleen
 - inferior to diaphragm, dorsolateral to stomach

Lymph Nodes

- Bean-shaped organs, up to 1 inch long, located along lymphatic vessels
 - scattered throughout body but concentrated near mammary glands, axillae & groin
- Stroma is capsule, trabeculae & reticular fibers
- Parenchyma is divided into 2 regions:
 - cortex
 - lymphatic nodules with germinal centers containing dendritic cells
 - antigen-presenting cells and macrophages
 - B cells proliferate into antibody-secreting plasma cells
 - medulla
 - contains B cells & plasma cells in medullary cords

Lymph Nodes



- Flow is in one direction
 - afferent vessels lead in
 - sinuses lead to efferent vessels that exit at hilus

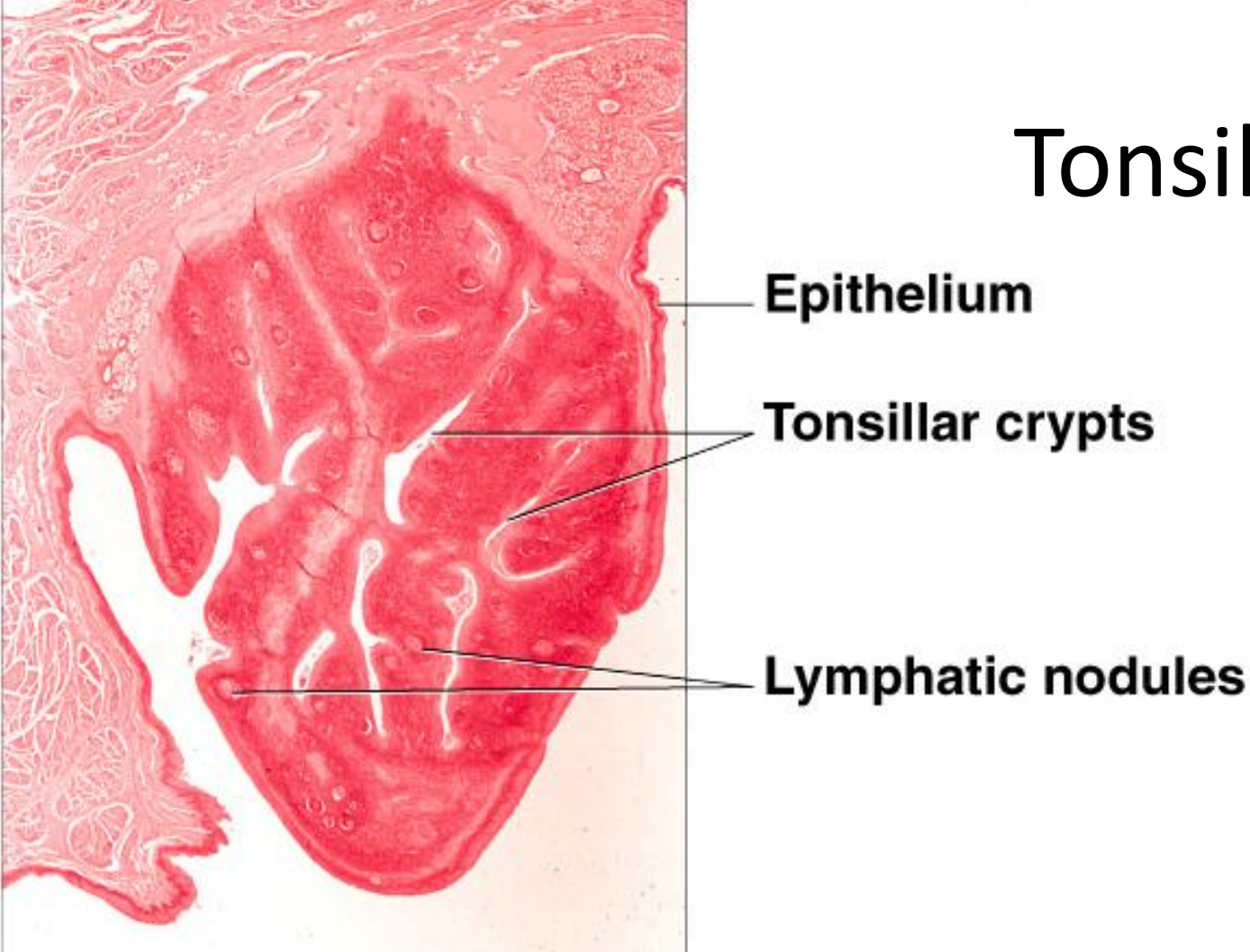
Metastasis Through Lymphatic System

- Characteristic of malignant tumors
- Spread of disease from one organ to another
 - cancer cells travel via blood or lymphatic system
 - cells establish new tumors where lodge
- Secondary tumor sites can be predicted by direction of lymphatic flow from primary site
- Cancerous lymph nodes are firm, enlarged and nontender -- infected lymph nodes are not firm and are very tender

Location of Tonsils

- **Palatine** tonsils
 - pair at the posterior margin of oral cavity; most often infected
- **Lingual** tonsils
 - pair at the root of tongue
- **Pharyngeal** tonsil (adenoid)
 - single tonsil on wall of pharynx

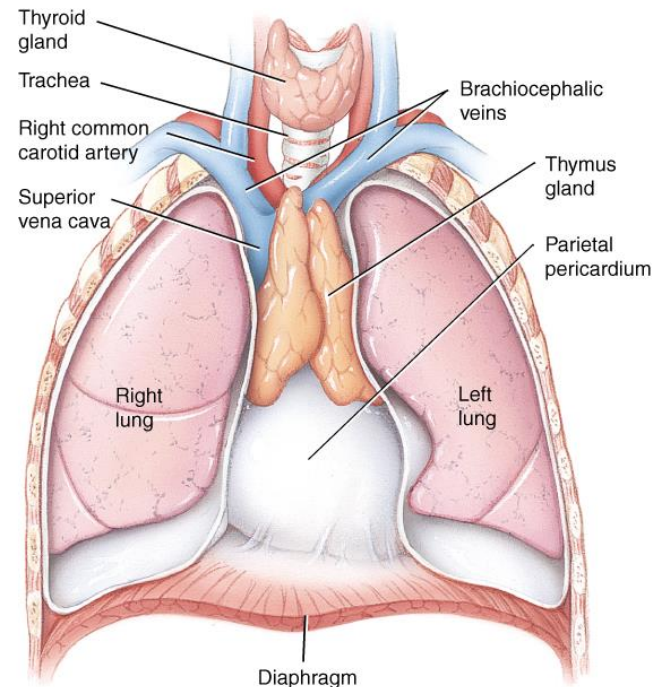
Tonsil



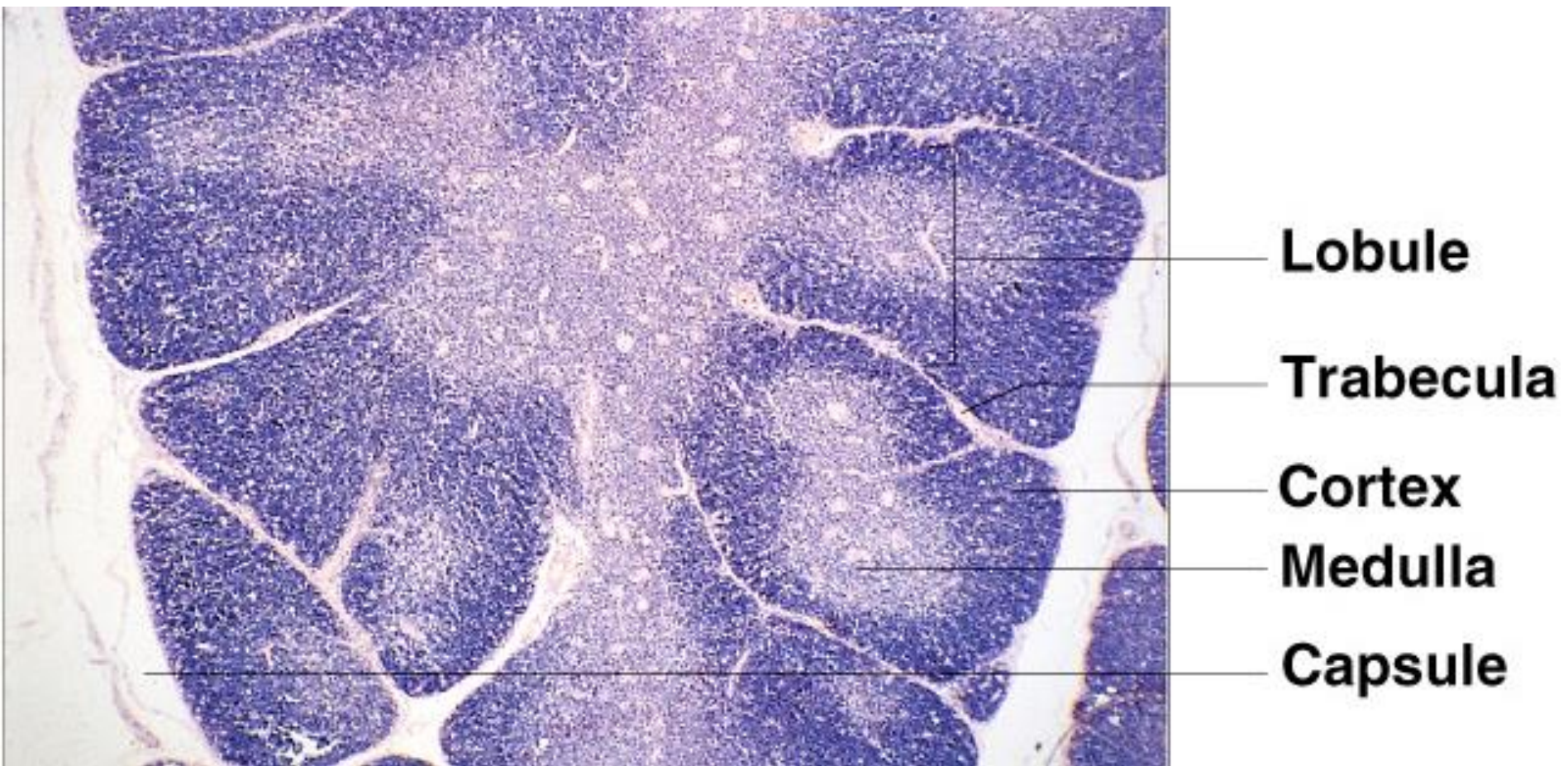
- Covered by epithelium
- Pathogens get into tonsillar crypts and encounter lymphocytes

Thymus Gland

- Large organ in infants (70 g) but atrophied as adult (3 g)
- 2 lobed organ located in mediastinum
- Capsule & trabeculae divide it into lobules
- Each lobule has cortex & medulla
 - tightly packed lymphocytes & macrophages
- Medulla
 - reticular epithelial cells produces thymic hormones
 - Hassall's corpuscles



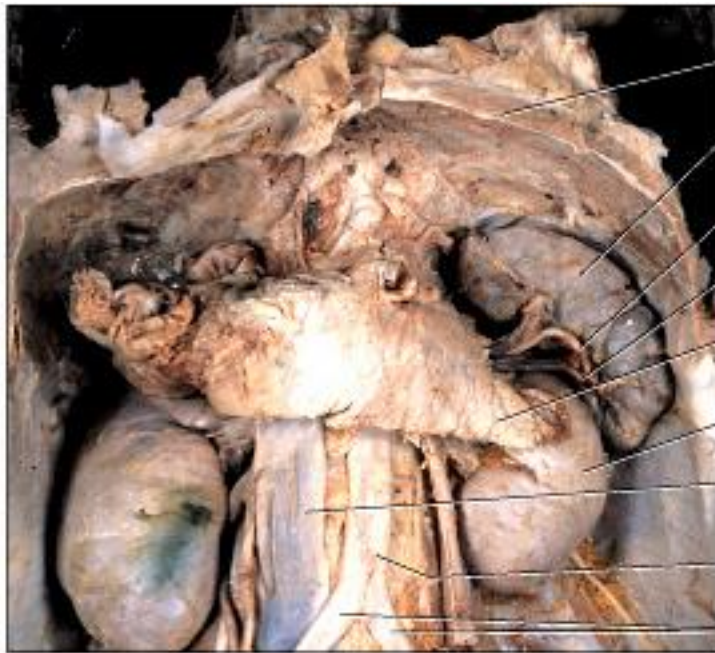
Histology of Thymus



Spleen

- Parenchyma appears in fresh specimens as
 - **red pulp**: sinuses filled with erythrocytes
 - **white pulp**: lymphocytes, macrophages; surrounds small branches of splenic artery
- Functions
 - blood production in fetus
 - blood reservoir
 - RBC disposal
 - immune reactions: **filters blood**, quick to detect antigens

Spleen



Diaphragm (cut)

Spleen

Splenic artery

Splenic vein

Pancreas

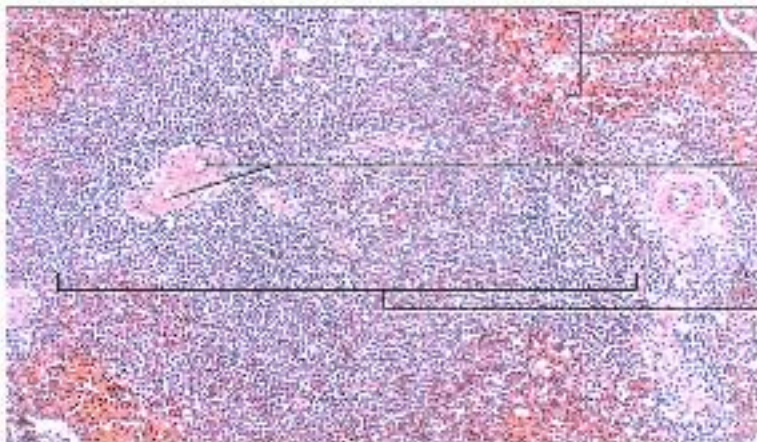
Kidney

Inferior vena cava

Aorta

Common iliac arteries

(a)



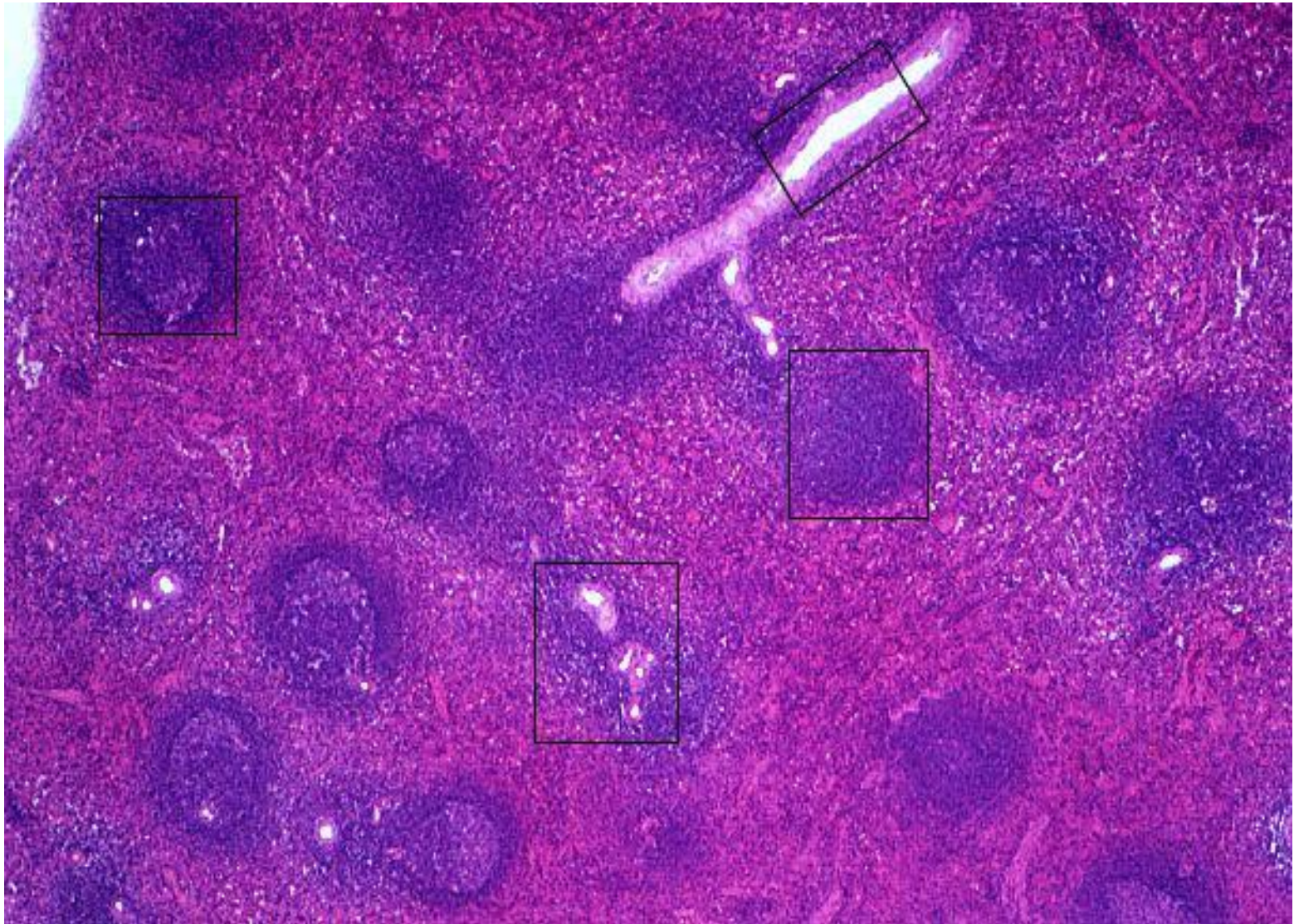
Red pulp

Central artery (branching)

White pulp

(b)

Spleen



Immunology

- Resistance is the ability to ward off disease
 - lack of resistance is termed susceptibility
- Nonspecific resistance to disease
 - general defensive mechanisms effective on a wide range of pathogens (disease producing microbes)
- Specific resistance or immunity is ability to fight a specific pathogen
 - cell-mediated immunity
 - antibody-mediated immunity

Nonspecific Resistance to Disease

- Immediate protection against wide variety of pathogens & foreign substances
 - lacks specific responses to specific invaders
- Mechanisms function regardless of type of invader
 - external mechanical & chemical barriers
 - internal nonspecific defenses
 - antimicrobial proteins
 - natural killer cells & phagocytes
 - inflammation & fever

Skin & Mucous Membranes

- Mechanical protection
 - skin (epidermis) closely packed, keratinized cells
 - shedding helps remove microbes
 - mucous membrane secretes viscous mucous
 - cilia & mucus trap & move microbes toward throat
 - washing action of tears, urine and saliva
- Chemical protection
 - sebum inhibits growth bacteria & fungus
 - perspiration lysozymes breakdown bacterial cells
 - acidic pH of gastric juice and vaginal secretions destroys bacteria

Leukocytes and Cutaneous Defenses

- Neutrophils
 - phagocytize bacteria
 - create a killing zone
 - degranulation: lysosomes discharge into tissue fluid, triggers
 - respiratory burst: toxic chemicals are created ($O_2^{\cdot-}$, H_2O_2 , $HClO$)
- Eosinophils
 - phagocytize antigen-antibody complexes, allergens, inflammatory chemicals
 - antiparasitic effects: aggregate and release enzymes

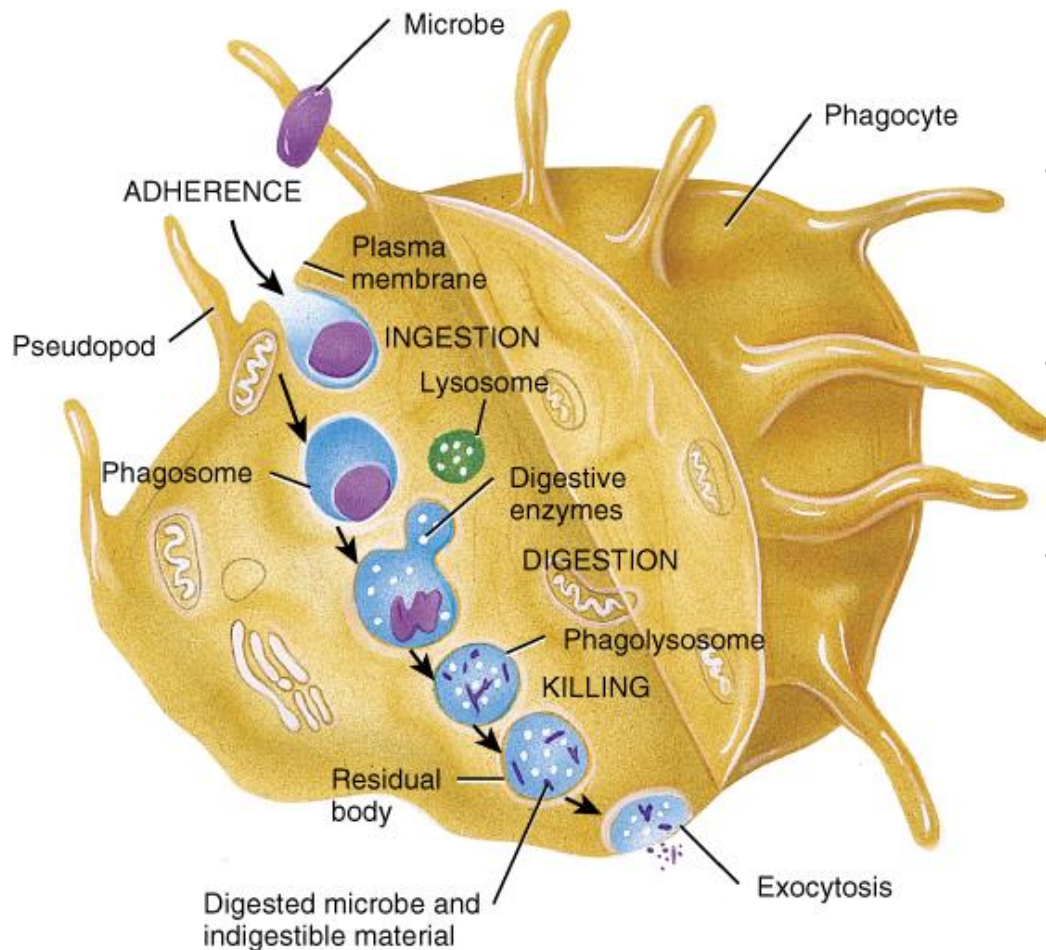
Internal Defenses

- Antimicrobial proteins discourage microbial growth
 - interferons
 - produced by virally infected lymphocytes & macrophages
 - diffuse to neighboring cells to induce synthesis of antiviral proteins
 - complement proteins
 - inactive proteins in blood plasma
 - when activated enhance immune, allergic & inflammatory reactions
 - transferrins
 - iron-binding proteins inhibit bacterial growth by reducing available iron

Natural Killer Cells & Phagocytes

- NK cells kill a variety of microbes & tumor cells
 - found in blood, spleen, lymph nodes & red marrow
 - attack cells displaying abnormal MHC antigens
- Phagocytes (neutrophils & macrophages)
 - ingest microbes or particulate matter
 - macrophages developed from monocytes
 - fixed macrophages stand guard in specific tissues
 - histiocytes in the skin, kupffer cells in the liver, alveolar macrophages in the lungs, microglia in the brain & macrophages in spleen, red marrow & lymph nodes
 - wandering macrophages in most tissue

Phagocytosis



- Chemotaxis
 - attraction to chemicals from damaged tissues, complement proteins, or microbial products
- Adherence
 - attachment to plasma membrane of phagocyte
- Ingestion
 - engulf by pseudopods to form phagosome
- Digestion & killing
 - merge with lysosome containing digestive enzymes & form lethal oxidants
 - exocytosis residual body

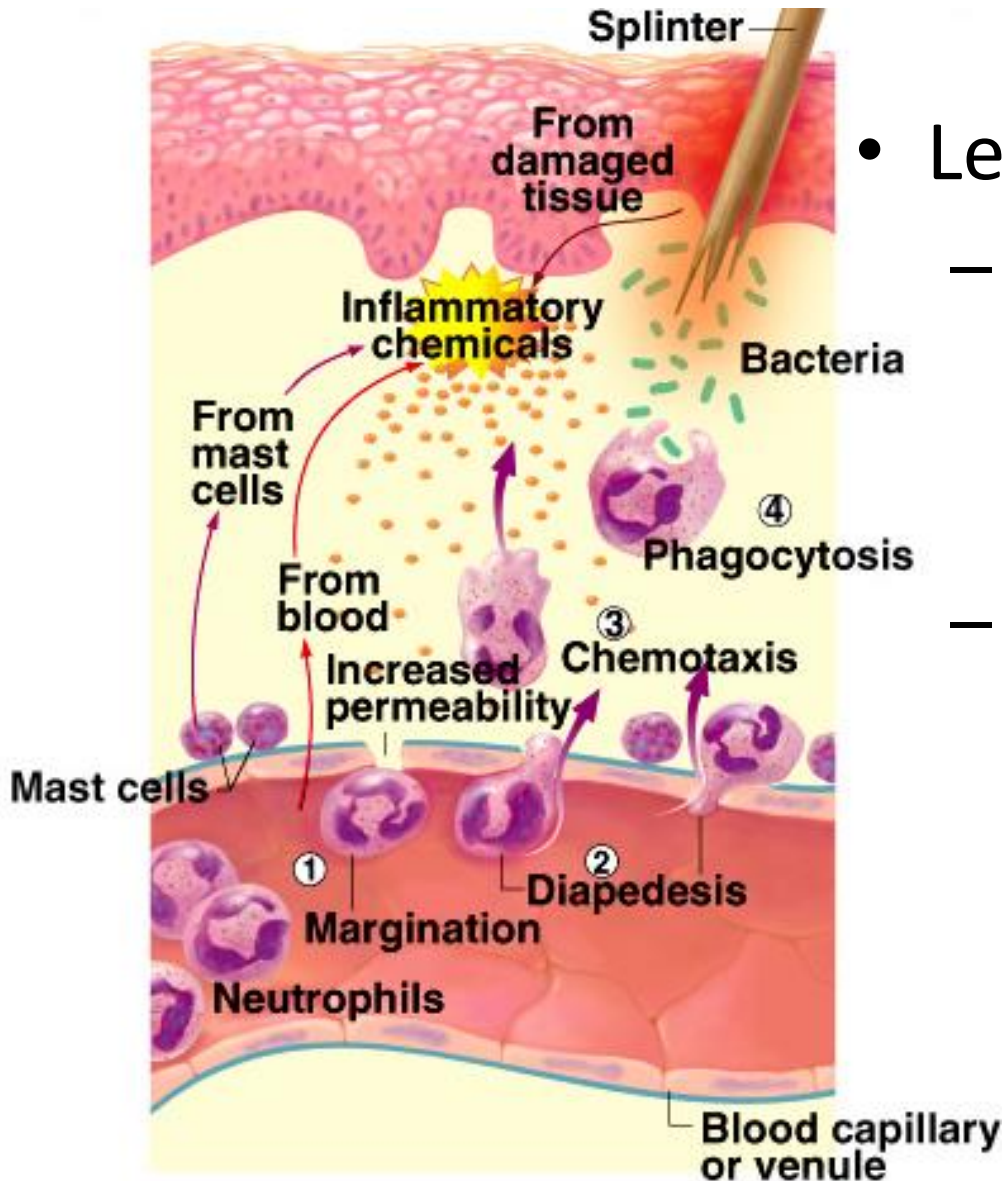
Inflammation

- Defensive response to tissue injury
 - limits spread of pathogens, then destroys them; removes debris, initiates tissue repair
 - suffix *-itis* denotes inflammation of specific organs
- Cardinal signs
 - **redness** (erythema) caused by hyperemia (↑ blood flow)
 - **swelling** (edema) caused by ↑ capillary permeability and filtration
 - **heat** caused by hyperemia
 - **pain** caused by inflammatory chemicals (bradykinin, prostaglandins) secreted by damaged cells, pressure on nerves

Mobilization of Defenses

- Bradykinin, histamine and leukotrienes are secreted by damaged cells, basophils (blood) and mast cells (tissue)
 - stimulates vasodilation that leads to **hyperemia**
 - causes redness and heat: $\uparrow$ **local metabolic rate**, promotes cell multiplication and healing
 - dilutes toxins, provides O_2 , nutrients, waste removal
 - stimulates $\uparrow$ permeability of blood capillaries
 - allows **blood cells**, plasma **chemicals** (antibodies, complement proteins, fibrinogen) **into tissue**
 - **clotting** sequesters bacteria, forms scaffold for tissue repair

Mobilization of Defenses



- Leukocyte Deployment
 - margination
 - leukocytes adhere to blood vessel walls (cell adhesion molecules)
 - diapedesis
 - leukocytes squeeze between endothelial cells into tissue space

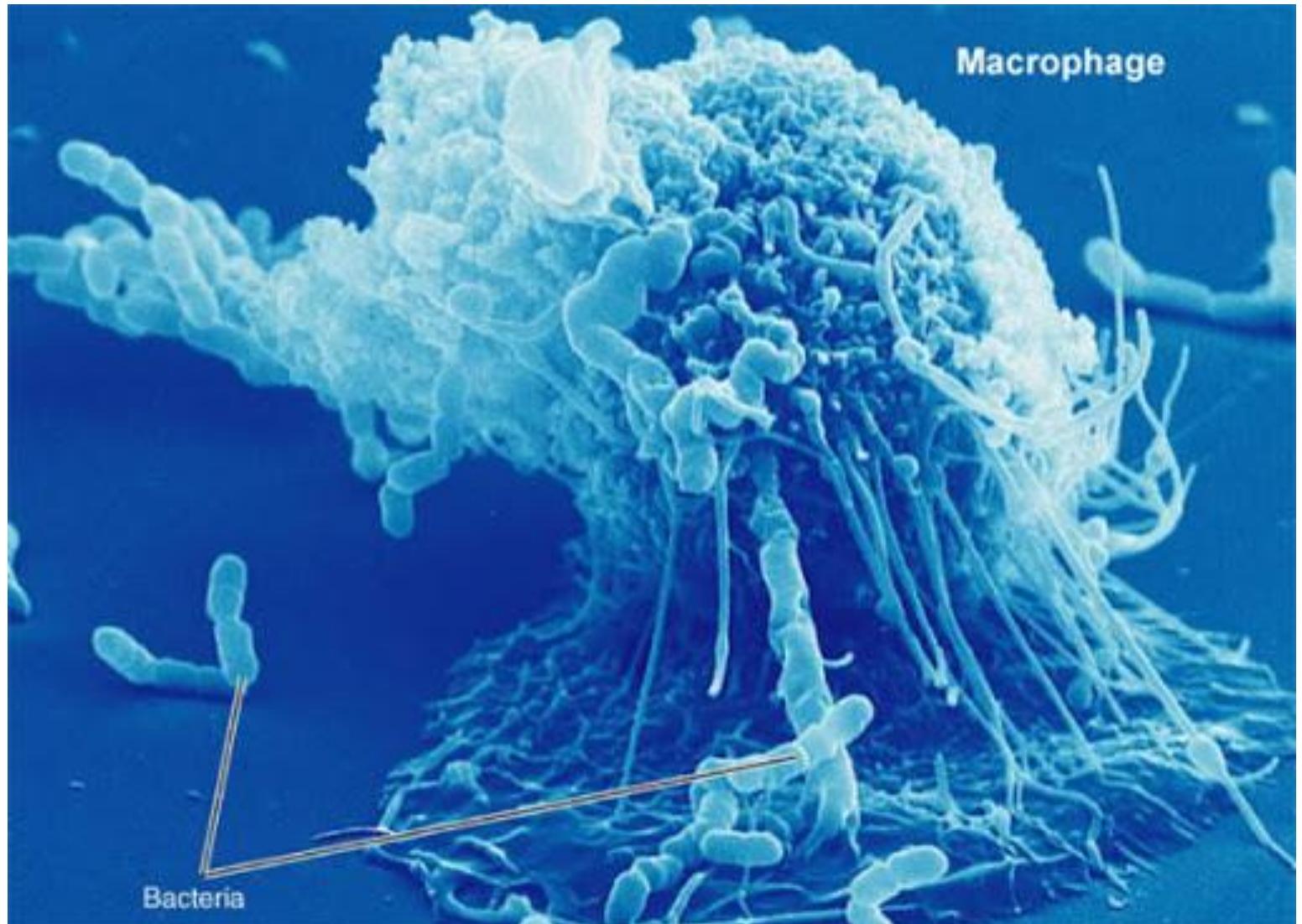
Containment and Destruction of Pathogens

- Fibrinogen now in tissue clots, trapping microbes
- Heparin prevents clotting at site of injury
 - pathogens trapped in a fluid pocket surrounded by clot
- Chemotaxis
 - leukocytes are attracted to inflammatory chemicals
- Neutrophils are quickest to respond
 - phagocytosis
 - respiratory burst
 - secrete cytokines for recruitment of macrophages and T cells

Tissue Cleanup

- Monocytes the primary agents of cleanup arrive in 8 to 12 hours, become macrophages,
- Edema ↓ venous flow, ↑ **lymphatic flow** that favors removal of bacteria and debris
- Formation of pus
 - mixture of tissue fluid, cellular debris, dying neutrophils and microbes

Macrophage adherence



Tissue Repair

- Blood platelets and endothelial cells in injured area secrete a cytokine, PDGF, that stimulates fibroblasts to multiply and synthesize collagen
- Facilitated by hyperemia that provides materials needed and heat that increases metabolism
- Fibrin clot may provide a scaffold for repair
- Pain limits use of body part allowing for repair

Abscesses and Ulcers

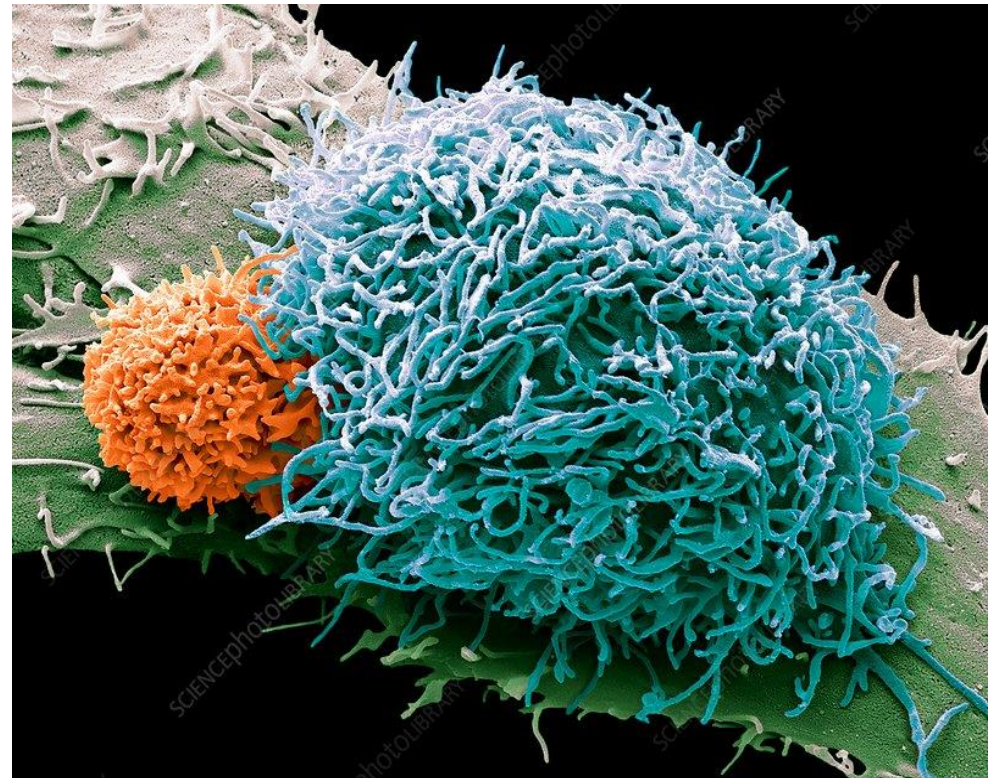
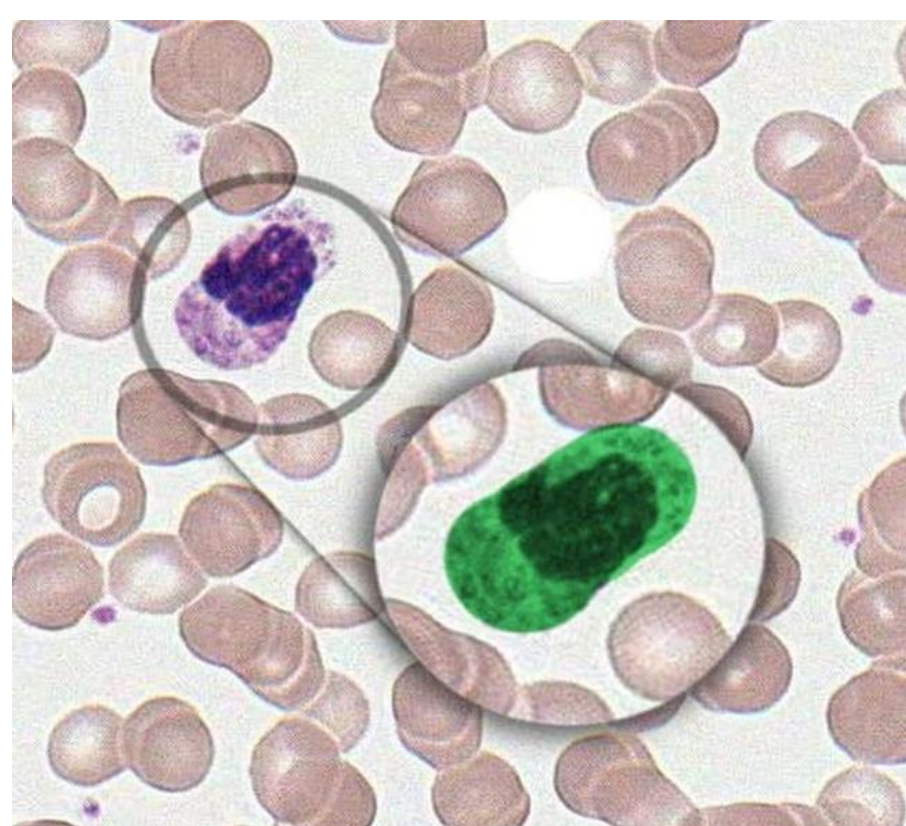
- Pus is dead phagocytes, damaged tissue cells & fluid
- Abscess is accumulation of pus in a confined space not open to the outside
 - pimples & boils
- Ulcer is an open sore
- People with poor circulation (diabetics with advanced atherosclerosis)
 - stasis ulcers in tissues of legs due to poor oxygen & nutrient supply to tissues

Abscess vs Ulcer

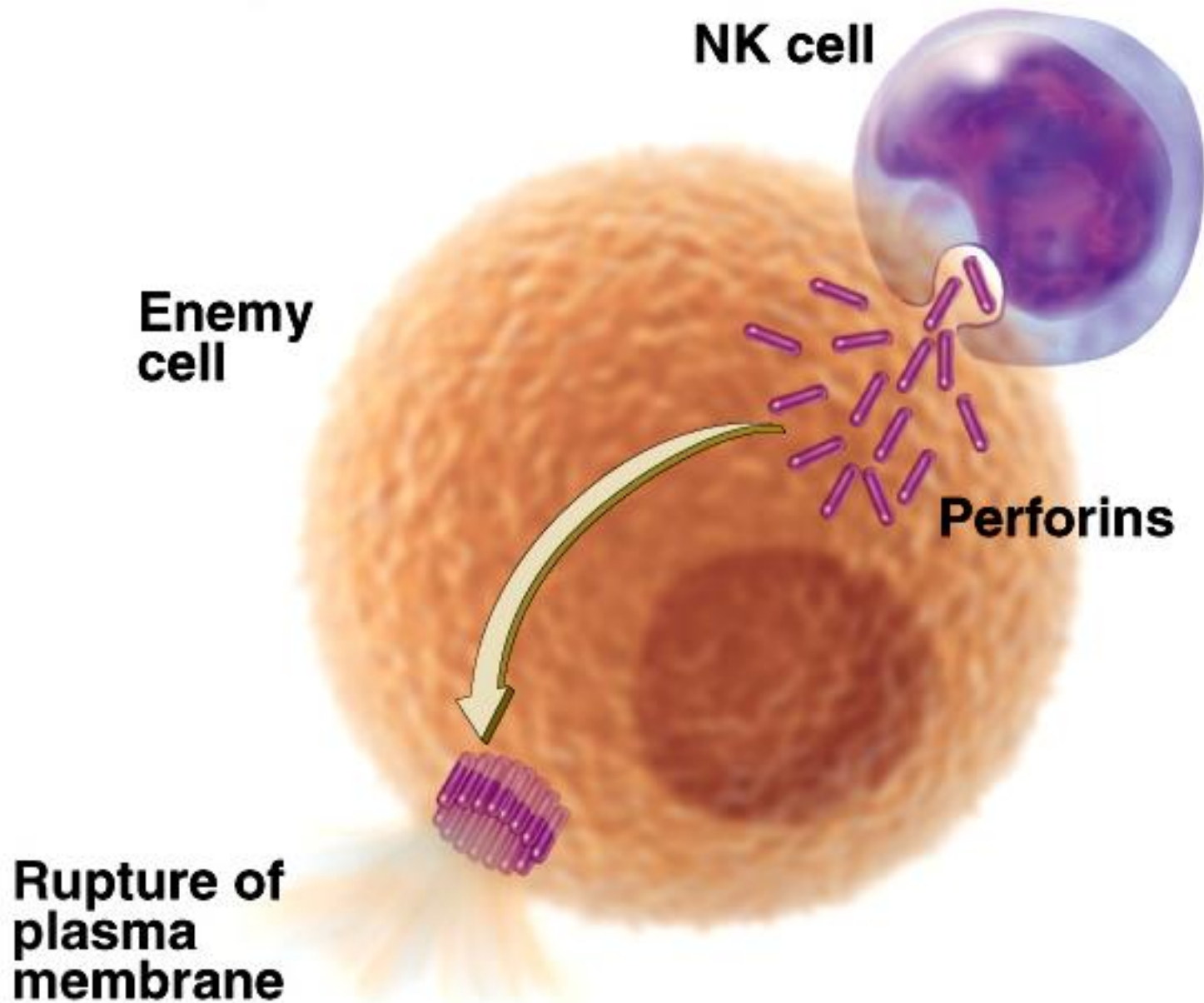


Natural Killer Cells

- Derived from T-lymphocyte precursor
- Specialized for self from funky self



Action of Perforin



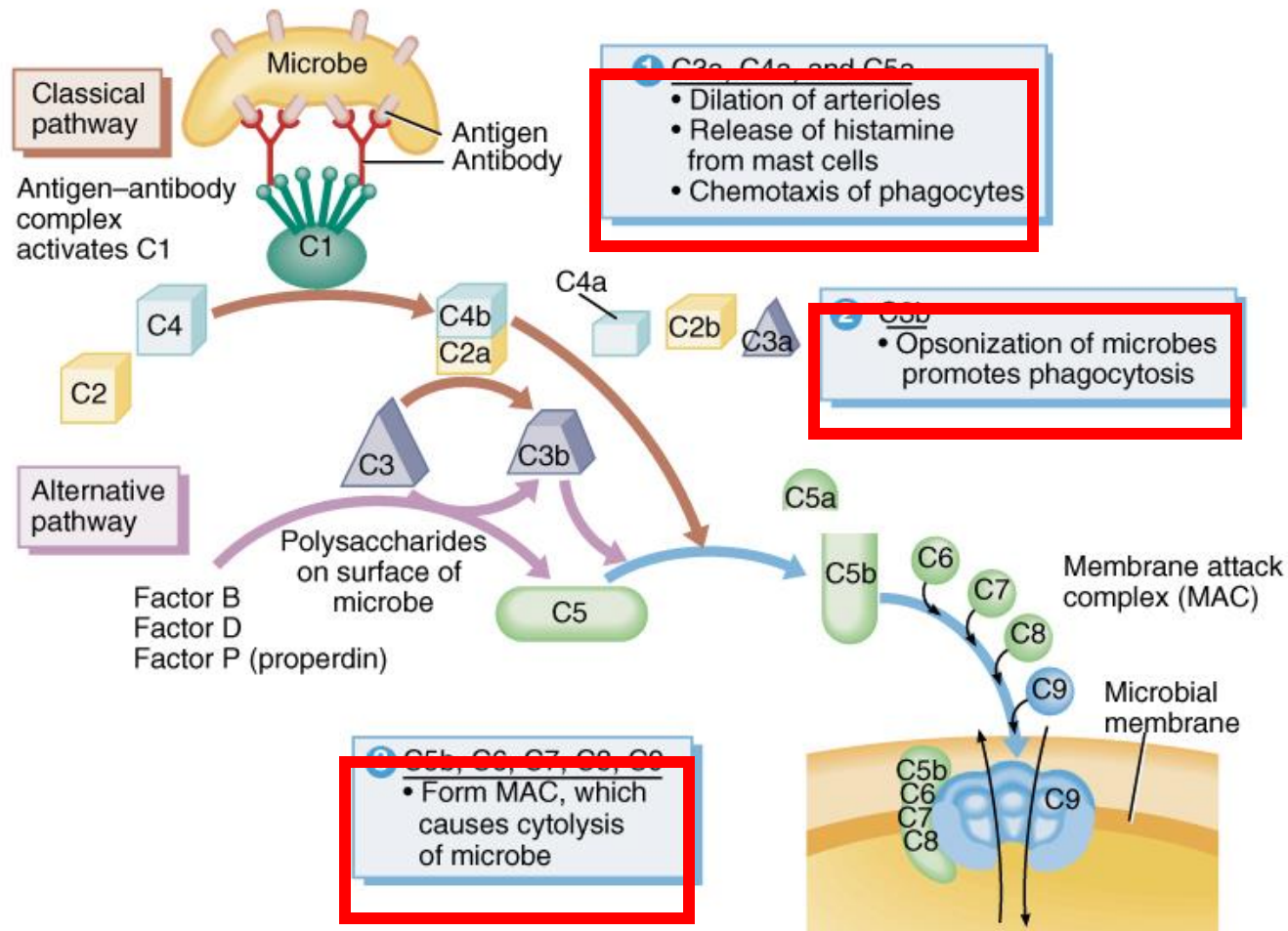
Antimicrobial Proteins

- Interferons: polypeptides secreted by cells invaded by viruses
 - antiviral effect
 - generalized protection
 - interferons diffuse to neighboring cells and stimulate them to produce antiviral proteins
 - activate natural killer cells and macrophages
 - destroy infected host cells
 - anticancer effect
 - stimulate destruction of cancer cells

Complement System

- Group of proteins in blood that must be activated by pathogens to exert their effect
- Pathways of complement activation (*see next slide*)
 - classical pathway
 - alternate pathway
- Mechanisms of action (*see next slide*)
 - enhanced inflammation
 - opsonization (promotes phagocytosis)
 - cytolysis (membrane attack complex)

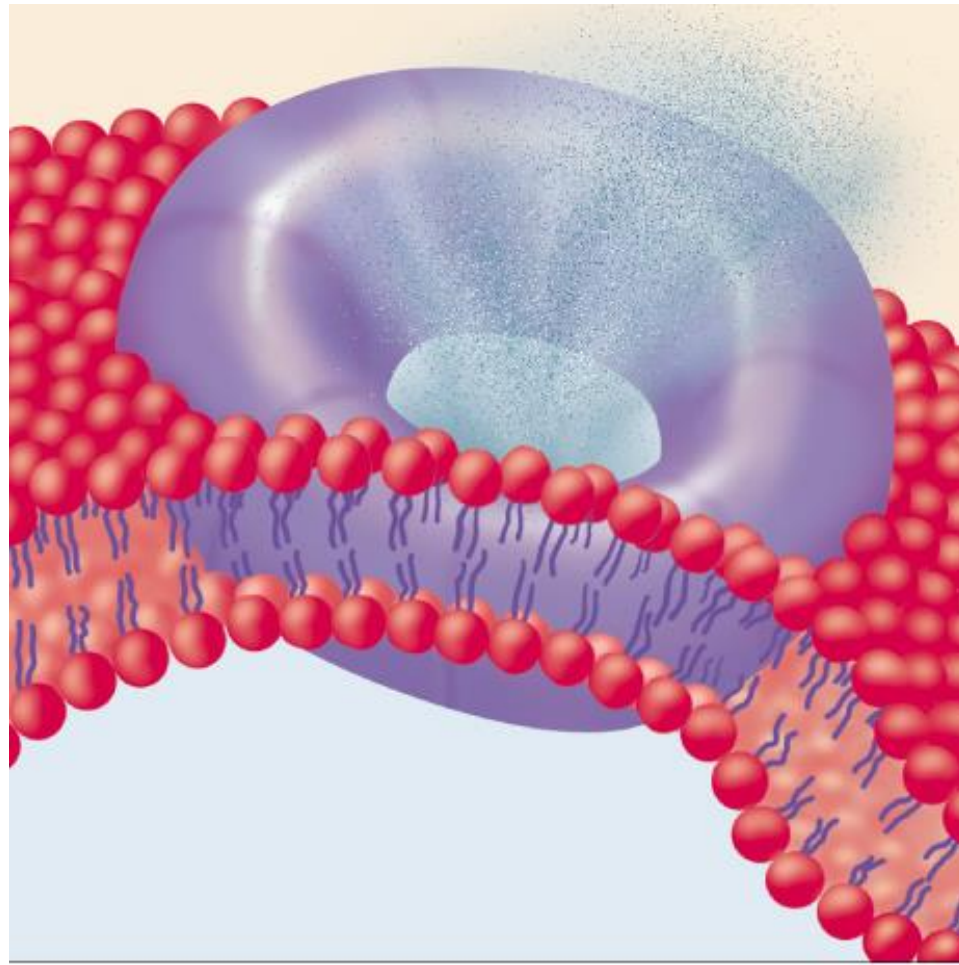
Pathways of the Complement System



- Classical pathway begins with activation of C1
- Alternate pathway begins with activation of C3
- Lead to inflammation, enhanced phagocytosis or microbe bursting

Membrane Attack Complex

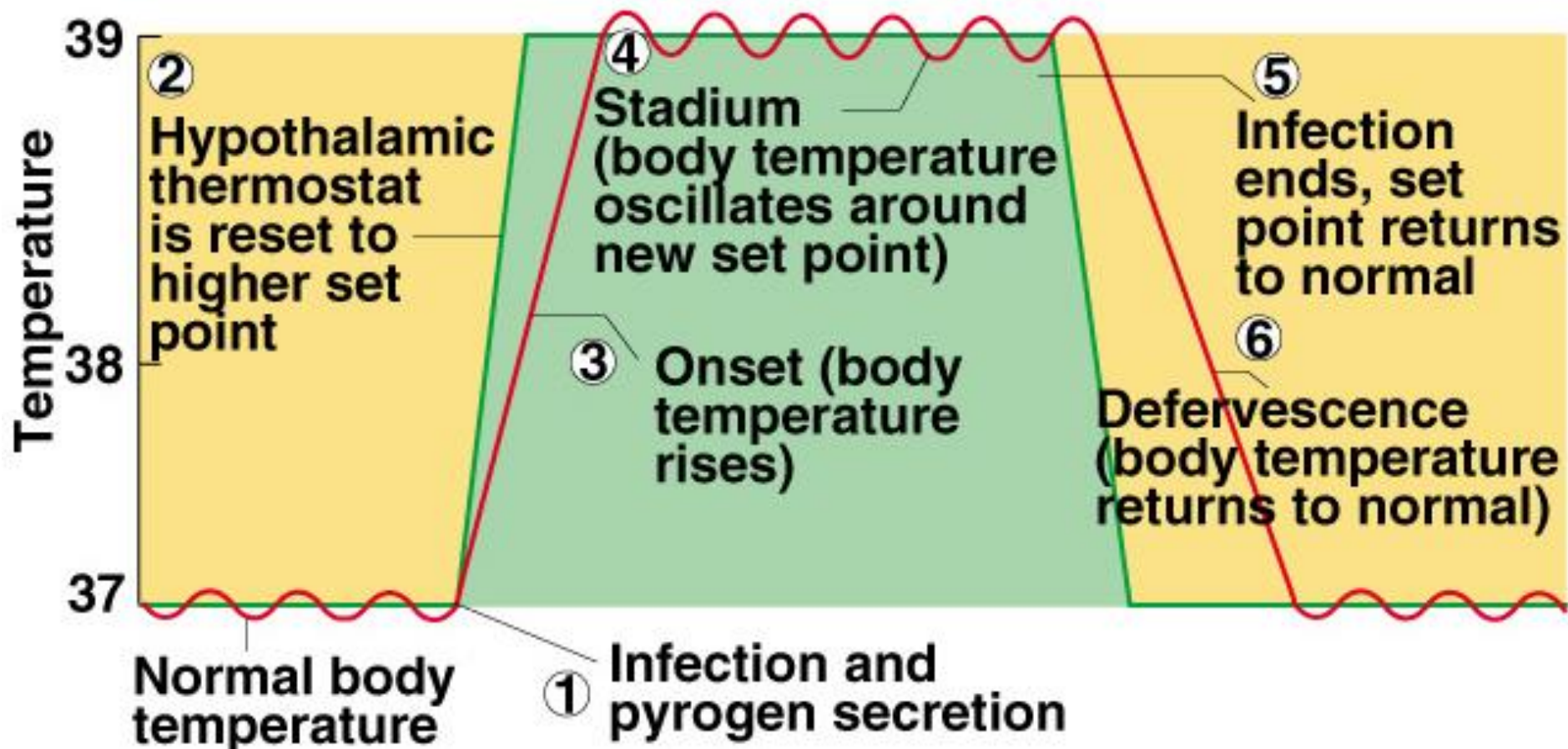
- Complement proteins C5b to C9 form ring in plasma membrane of enemy cell causing cytolysis



Fever

- Defense mechanism: can do more good than harm
 - promotes interferon activity
 - accelerating metabolic rate and tissue repair
 - inhibiting pathogen reproduction
- A cytokine, interleukin 1 is called a pyrogen
 - secreted by macrophages, stimulates anterior hypothalamus to secrete PGE which resets body thermostat higher
 - > 105°F may cause delirium, 111°F- 115°F, coma-death
- Stages of fever
 - onset, stadium, defervescence

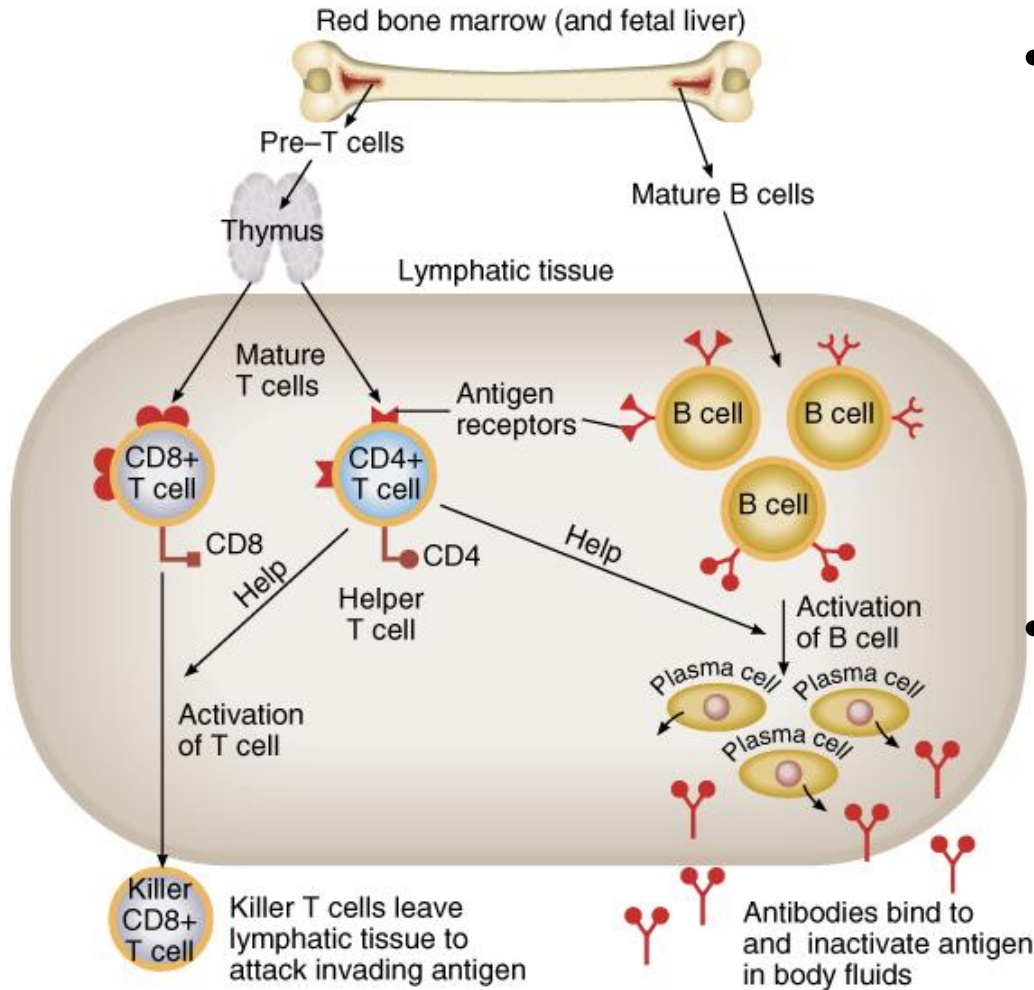
Course of a Fever



Specific Resistance: Immunity

- Immunity is bodies ability to defend itself against specific foreign material or organisms
 - bacteria, toxins, viruses, cat dander, etc.
- Differs from nonspecific defense mechanisms
 - specificity----recognize self & non-self
 - memory----2nd encounter produces even more vigorous response
- Immune system is cells and tissues that produce the immune response
- Immunology is the study of those responses

Maturation of T and B Cells



- T cell mature in thymus
 - cell-mediated response
 - killer cells attack antigens
 - helper cells costimulate T and B cells
 - effective against fungi, viruses, parasites, cancer, and tissue transplants
- B cells in bone marrow
 - antibody-mediated response
 - plasma cells form antibodies
 - effective against bacteria

CELL-MEDIATED IMMUNE RESPONSES
Directed against intracellular pathogens, such as viruses, some cancer cells, and tissue transplants

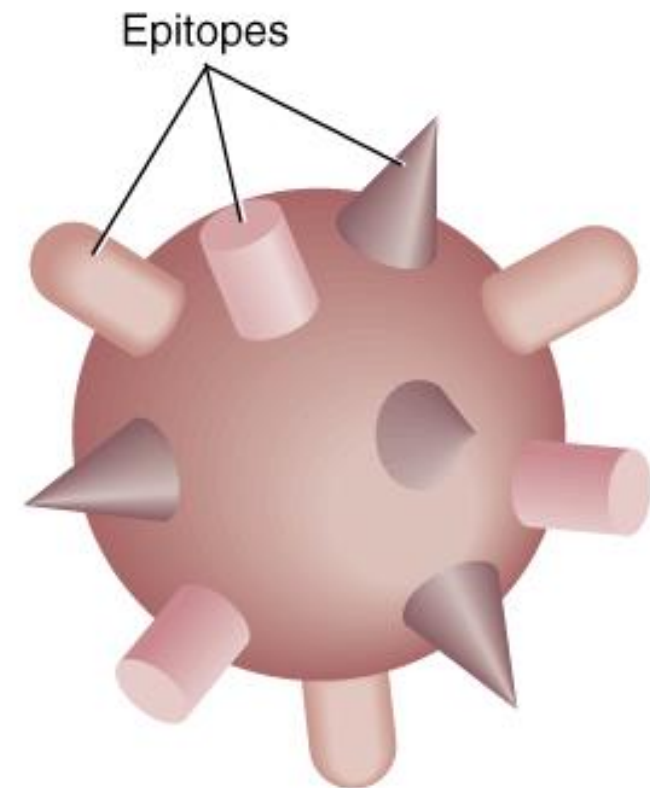
ANTIBODY-MEDIATED IMMUNE RESPONSES
Directed against extracellular pathogens, such as bacteria

Antigens

- Molecules or bits of foreign material
 - entire microbes, parts of microbes, bacterial toxins, pollen, transplanted organs, incompatible blood cells
- Required characteristics to be considered an antigen
 - immunogenicity = ability to provoke immune response
 - reactivity = ability to react to cells or antibodies it caused to be formed
- Get past the bodies nonspecific defenses
 - enter the bloodstream to be deposited in spleen
 - penetrate the skin & end up in lymph nodes
 - penetrate mucous membrane & lodge in associated lymphoid tissue

Chemical Nature of Antigens/Epitopes

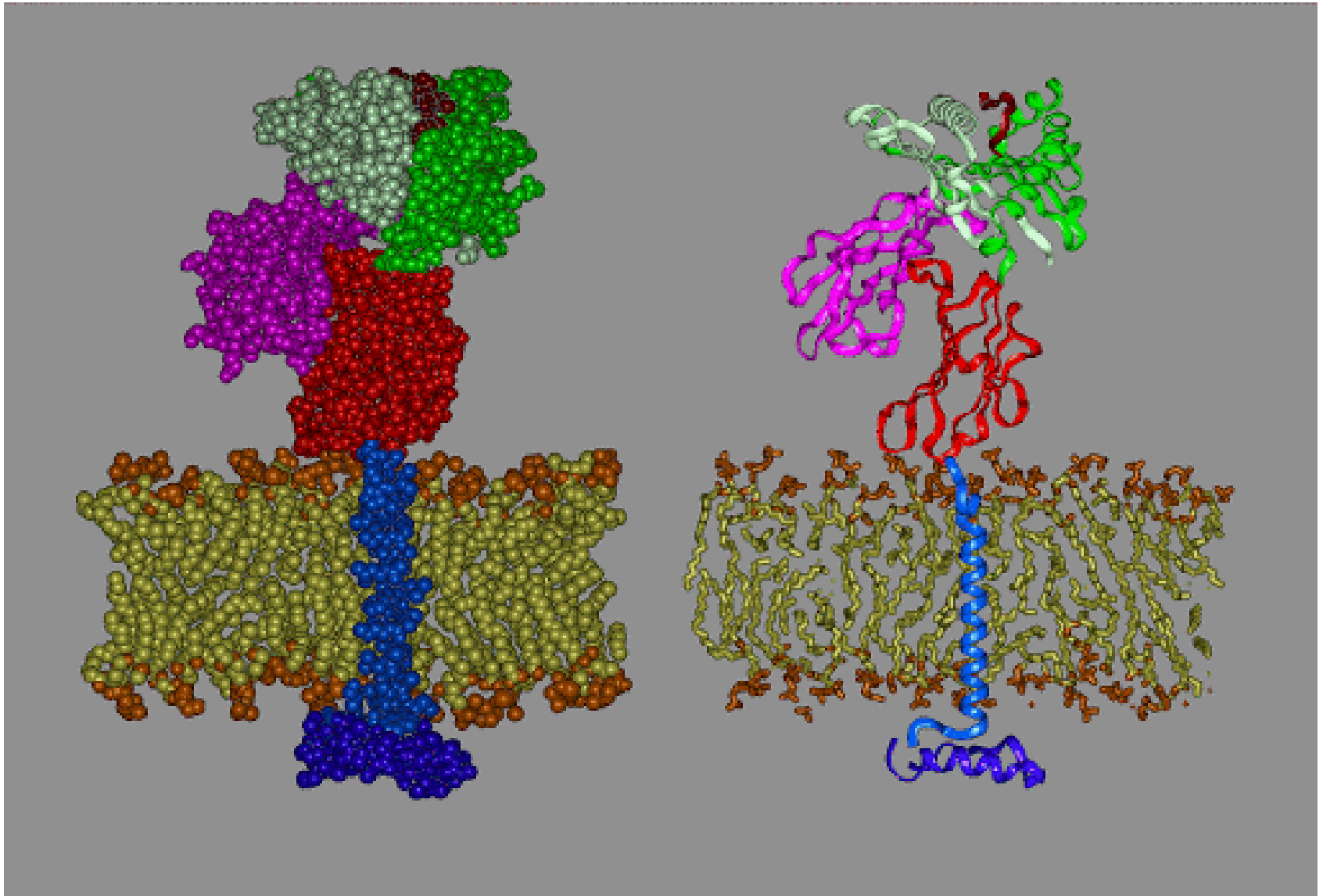
- Large, complex molecules, usually proteins
 - if have simple repeating subunits are not usually antigenic (plastics in joint replacements)
 - small part of antigen that triggers the immune response is epitope
 - antigenic determinant
- Hapten is smaller substance that can not trigger an immune response unless attached to body protein
 - lipid of poison ivy



Diversity of Antigen Receptors

- Immune system can recognize and respond to a billion different epitopes -- even artificially made molecules
- Explanation for great diversity of receptors is genetic recombination of few hundred small gene segments
- Each B or T cell has its own unique set of gene segments that codes its unique antigen receptor in the cell membrane

Major Histocompatibility Complex



Major Histocompatibility Complex Antigens

- All our cells have unique surface markers (1000s molecules)
 - integral membrane proteins called HLA antigens
- MHC-I molecules are built into cell membrane of all cells except red blood cells
- MHC-II markers seen only on membrane of antigen presenting cells (macrophages, B cells, thymus cells)
- Function
 - if cell is infected with virus MHC-I contain bits of virus marking cell so T cells recognize is problem
 - if antigen presenting cells (macrophages or B cells) ingest foreign proteins, they will display as part of their MHC-II

MHC class I

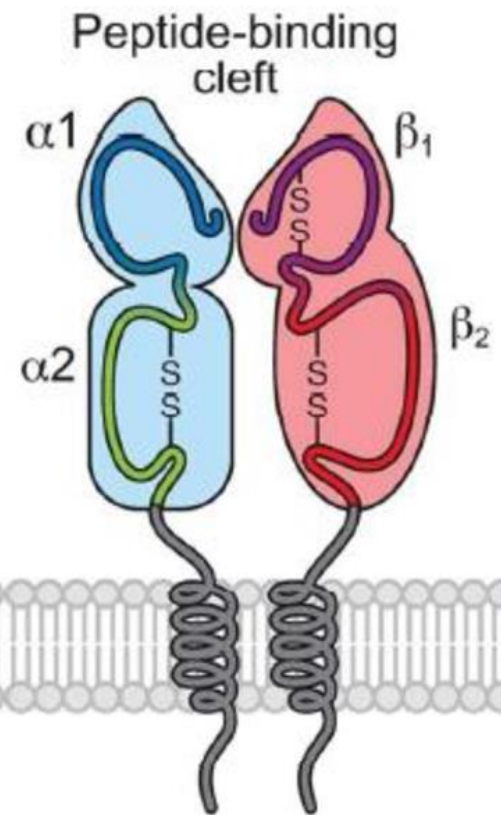
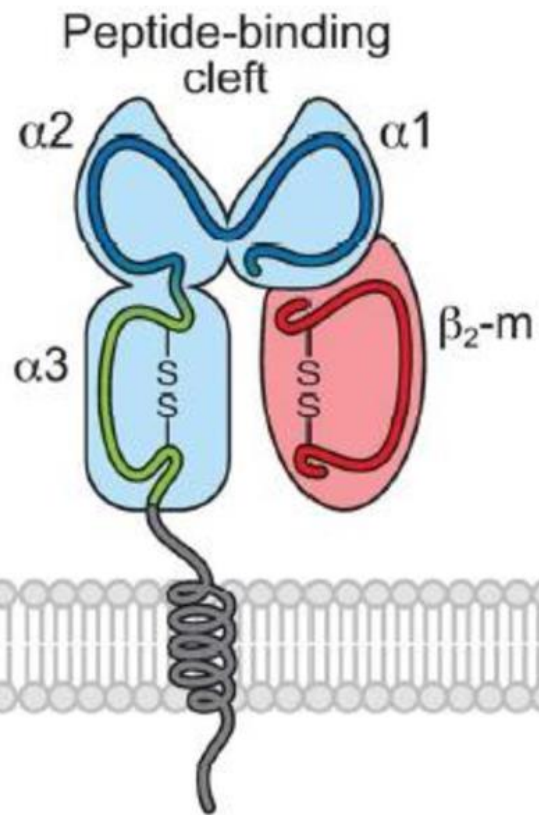
MHC class II

Membrane-distal domains

Membrane-proximal domains
(Ig-fold structure)

Transmembrane segment

Cytoplasmic tail



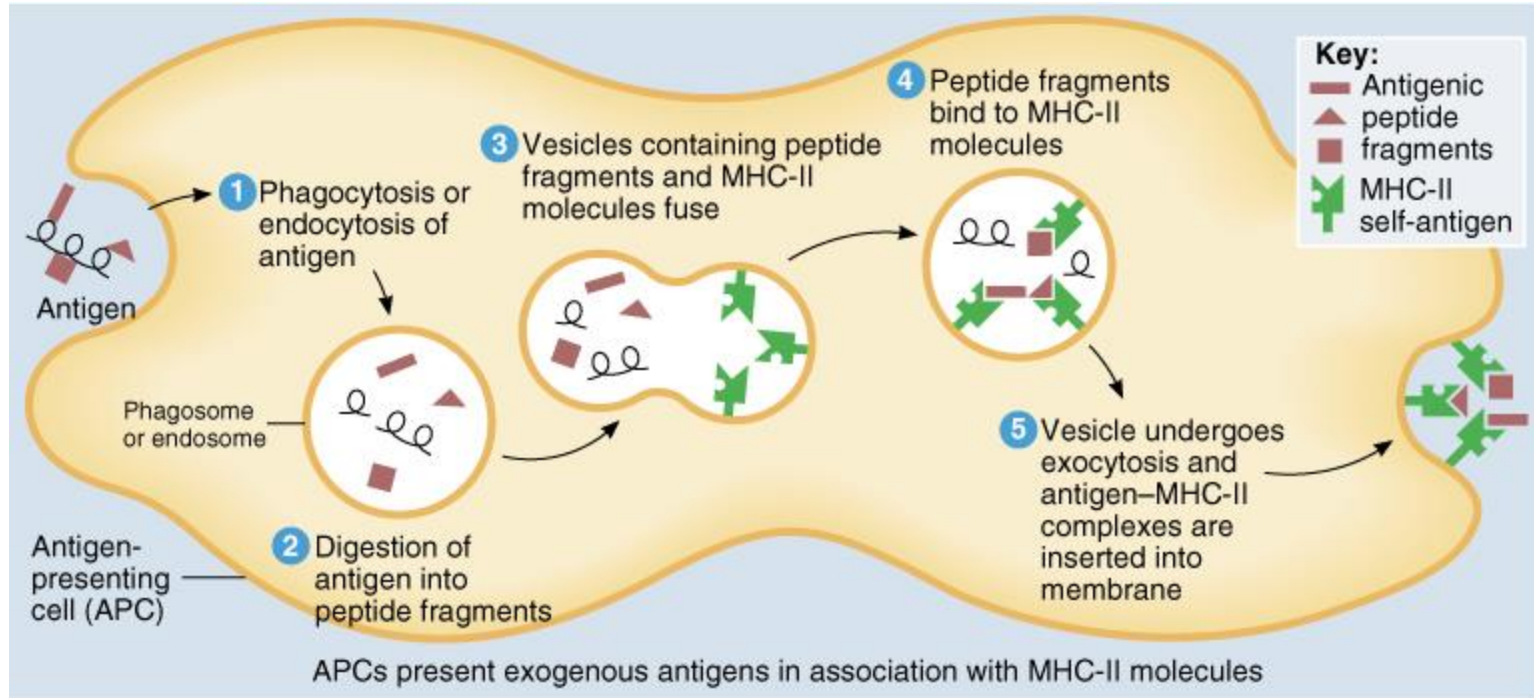
Histocompatibility Testing

- Histocompatibility is a similarity of MHC antigens on body cells of different individuals
 - tissue typing must be done before any organ transplant
 - can help identify biological parents

Pathways of Antigen Processing

- B and T cells must recognize a foreign antigen before beginning their immune response
 - B cells can bind to antigen in extracellular fluid
 - T cells can only recognize fragments of antigens that have been processed and presented to them as part of a MHC molecule
 - Helper T cells “see” antigens if part of MHC-II molecules on surface of antigen presenting cell
 - Cytotoxic T cells “see” antigens if part of MHC-I molecules on surface of body cells

Processing of Exogenous Antigens



- Foreign antigen in body fluid is phagocytized by APC
 - macrophage, B cell, dendritic cell (Langerhans cell in skin)
- Antigen is digested and fragments are bound to MHC-II molecules stuck into antigen presenting cell membrane
- APC migrates to lymphatic tissue to find T cells

Processing of Endogenous Antigens

- Endogenous antigens are foreign proteins produced within a body cell --- viral or cancerous
- Fragments of weird proteins become part of MHC-I molecules displayed at surface of cell
- Signals that a cell need help because it is infected or has turned cancerous

Cytokines & Cytokine Therapy

- Small protein hormones involved in immune responses
 - secreted by lymphocytes and antigen presenting cells
- Cytokine therapy uses cytokines (interferon)
 - alpha-interferon used to treat Kaposi's sarcoma, genital herpes, hepatitis B and C & some leukemias
 - beta-interferon used to treat multiple sclerosis
 - interleukin-2 used to treat cancer (side effects)

Lymphocytes

- Specific immunity depends on lymphocytes
- Types of lymphocytes in circulating blood
 - 80% **T cells** (cell mediated)
 - 15% **B cells** (antibody mediated)
 - 5% NK cells (nonspecific immunity)

T Lymphocytes (T cells)

- Stem cells of fetus colonize thymus for 2-3 days
- Thymosins stimulate maturing T cells to produce antigen receptors
 - **immunocompetent** cell: antigen receptors in place

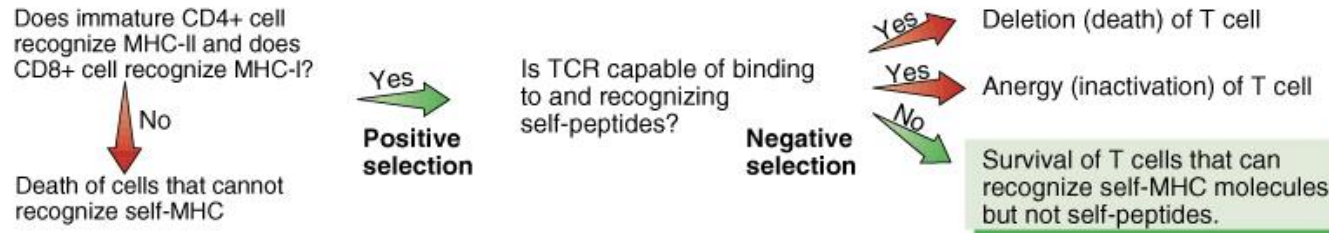
Negative Selection of T cells in Thymus

- Immunocompetent T cells must be able to
 1. bind to Reticular Epithelium cell
 2. not react to self antigens
- Failing either of these results in **negative selection** via
 - **clonal deletion**: destruction of offending T cell clones
 - **anergy**: inactive state, alive but unresponsive
 - leaves the body in a state of self-tolerance
- 1 in 100 emerge immunocompetent

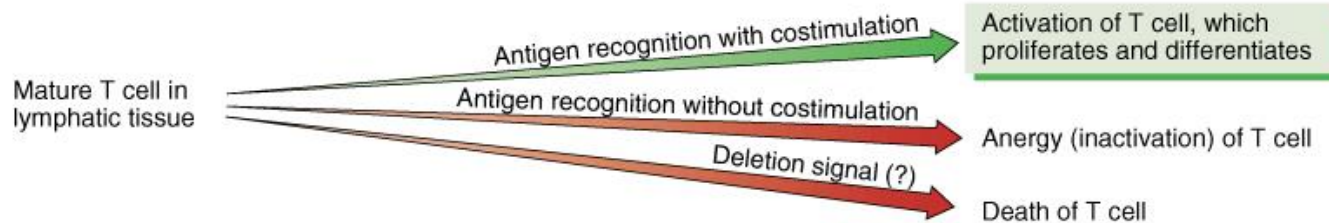
Positive Selection of T cells in Thymus

- Immunocompetent T cells that are able to
 1. bind to MHC on Reticular Epithelium cell
 2. not react to self antigensdivide rapidly and form a **clone** of T cells with identical receptors for antigen
- These cells leave thymus to colonize lymphatic tissues and organs

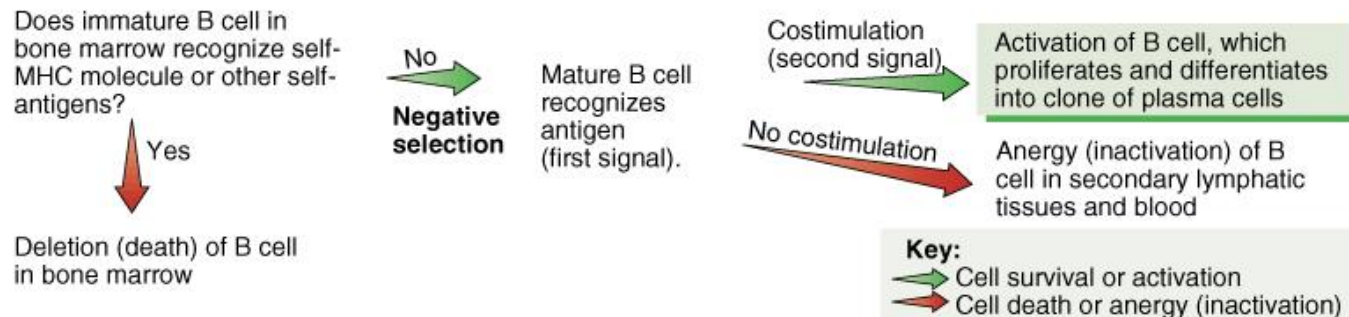
Development of Self-Recognition & Immunological Tolerance



(a) Positive and negative selection of T cells in the thymus gland



(b) Selection of T cells after they emerge from the thymus gland



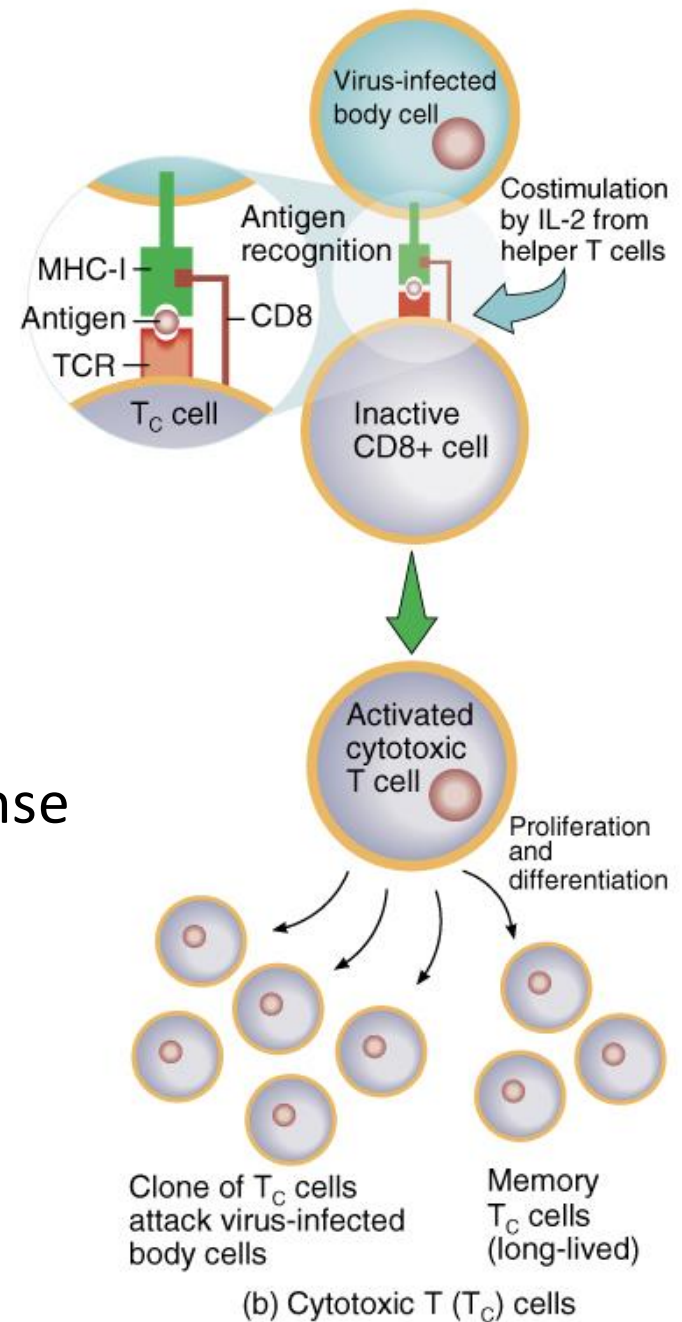
(c) Selection of B cells

Cellular Immunity

- T lymphocytes attack and destroy foreign cells and diseased host cells
- Involves 4 classes of T cells
 1. Cytotoxic (killer) T cells: carry out attack
 2. Helper T cells: help promote T cell and B cell action and nonspecific defense mechanisms
 3. Suppressor T cells: limit cell mediated attack
 4. Memory T cells: provide immunity from future exposure to antigen

Activation, Proliferation & Differentiation of Cytotoxic T Cells

- Receptor on CD8 cell binds to foreign antigen fragment part of MHC-I
- Costimulation from helper T cell
 - prevents accidental immune response
- Proliferates & differentiates into population (clone) of T_C cells and memory T_C cells
- Occurs in secondary lymphatic organs such as lymph node



T_C cell Recognition

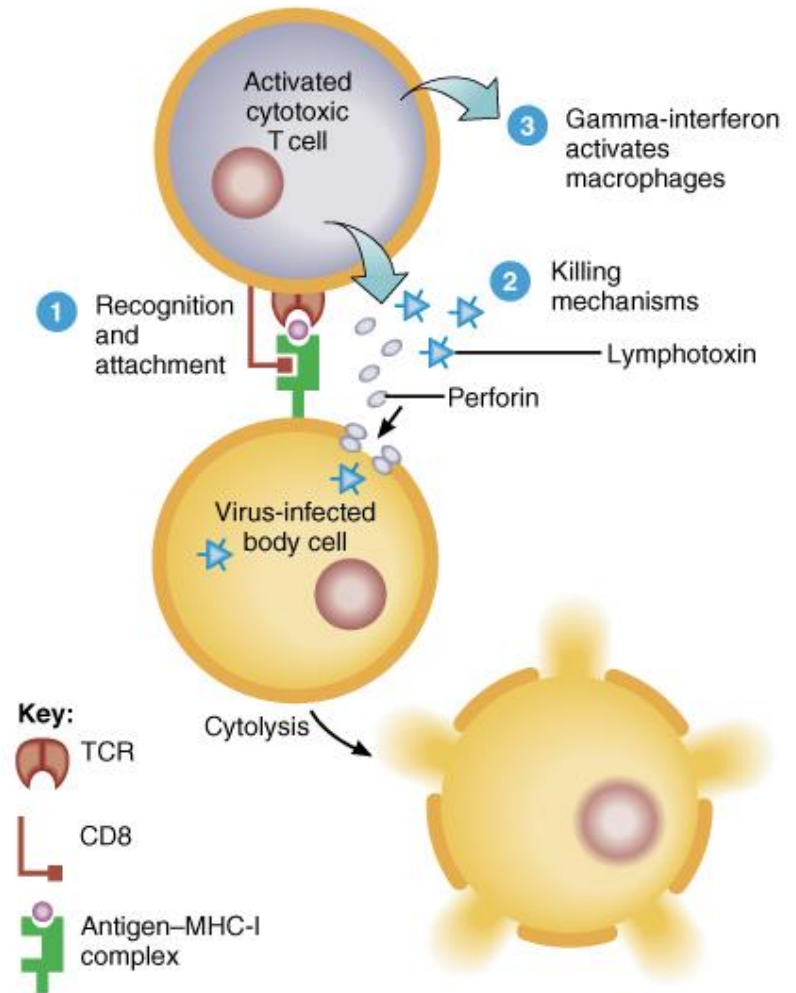
- Antigen presentation
 - **MHC-I** proteins
 - found on nearly all nucleated **body cells**
 - display peptides **produced by** host cells
- T_C cell activation
 1. binding of **cytotoxic T cells (CD8 cells)** to abnormal peptides on MHC-I and
 2. costimulation via a cytokine
 - triggers clonal selection: clone of identical T cells against cells with same epitope

Cytotoxic T Cells

- Display CD8 on surface
- Known as T8 or Tc or killer T cells
- Recognize antigen fragments associated with MHC-I molecules
 - cells infected with virus
 - tumor cells
 - tissue transplants
- Costimulation required by cytokine from helper T cell

Elimination of Invaders

- Cytotoxic T cells migrate to site of infection or tumor formation
- Recognize, attach & attack
 - secrete granules containing perforin that punch holes in target cell
 - secrete lymphotoxin that activates enzymes in the target cell causing its DNA to fragment
 - secrete gamma-interferon to activate phagocytic cells
 - Tumor necrosis factor kills cancer cells in 2-3 days



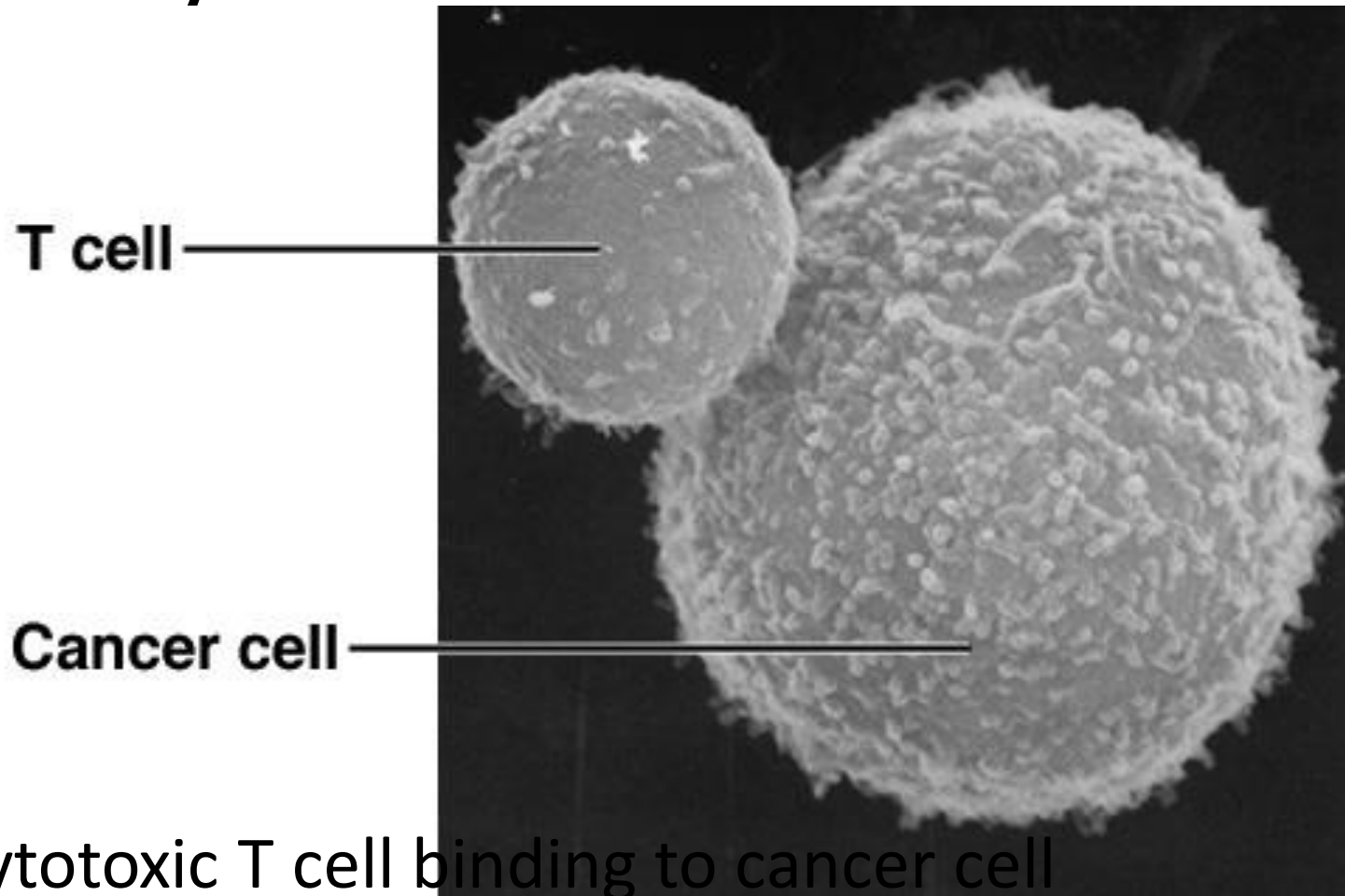
Immunological Surveillance

- Cancerous cell displays weird surface antigens (tumor antigens)
- Surveillance = immune system finds, recognizes & destroys cells with tumor antigens
 - done by cytotoxic T cells, macrophages & natural killer cells
 - most effective in finding tumors caused by viruses
- Transplant patients taking immunosuppressive drugs suffer most from viral-induced cancers

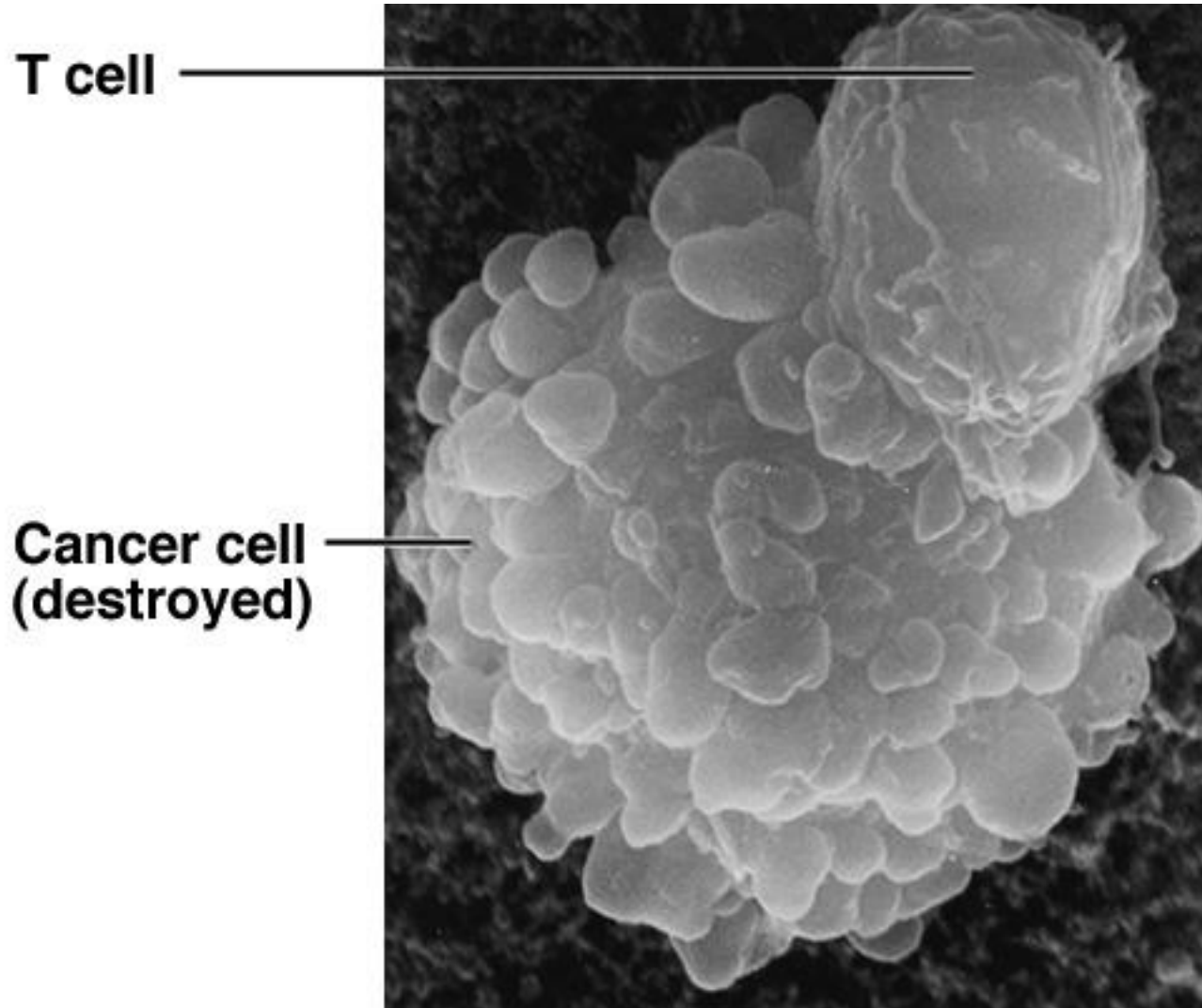
Graft Rejection

- After organ transplant, immune system has both cell-mediated and antibody-mediated immune response = graft rejection
- Close match of histocompatibility complex antigens has weaker graft rejection response
 - immunosuppressive drugs (cyclosporine)
 - inhibits secretion of interleukin-2 by helper T cells
 - little effect on B cells so maintains some resistance

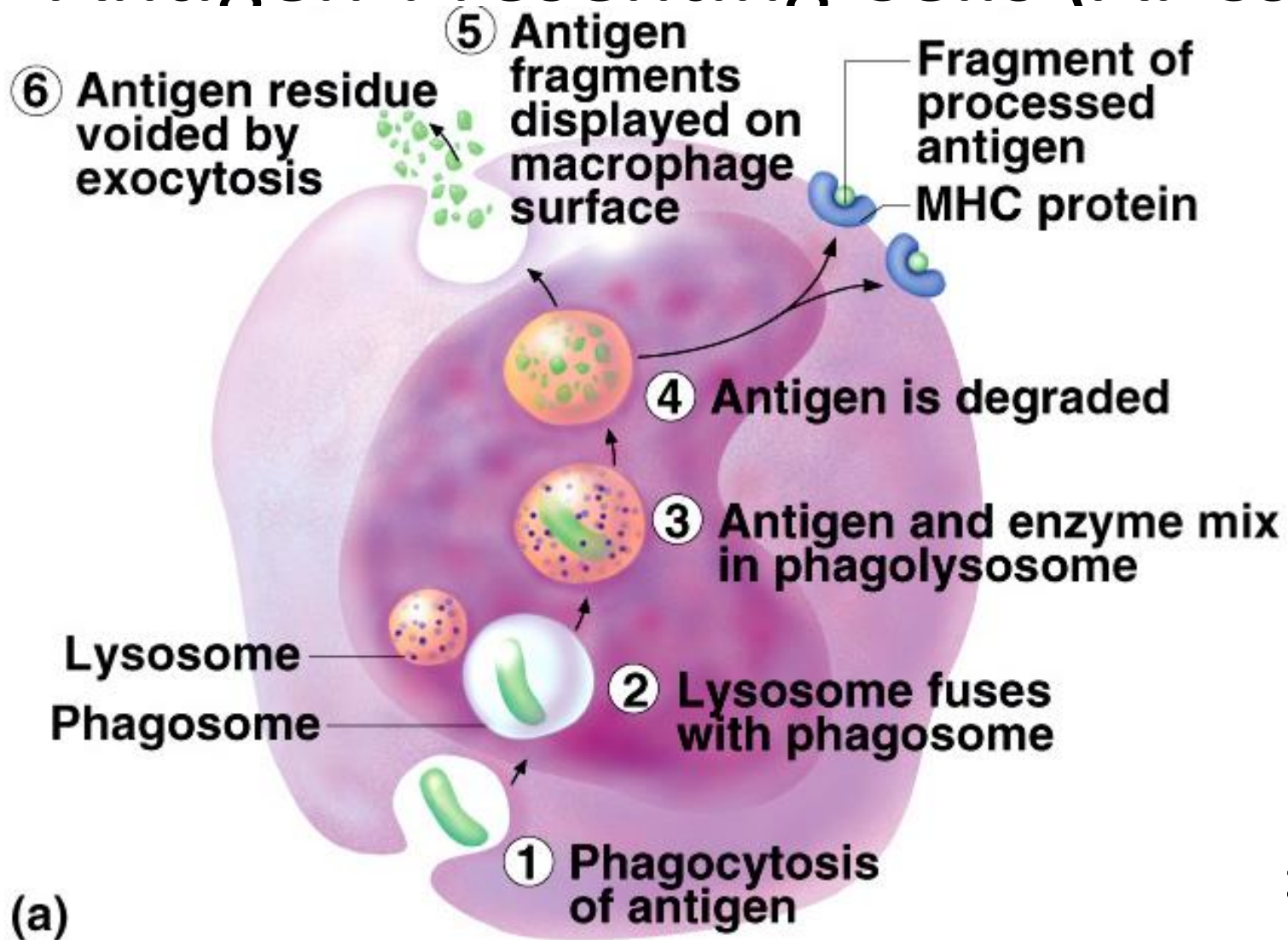
Cytotoxic T Cell Function



Destruction of Cancer Cell



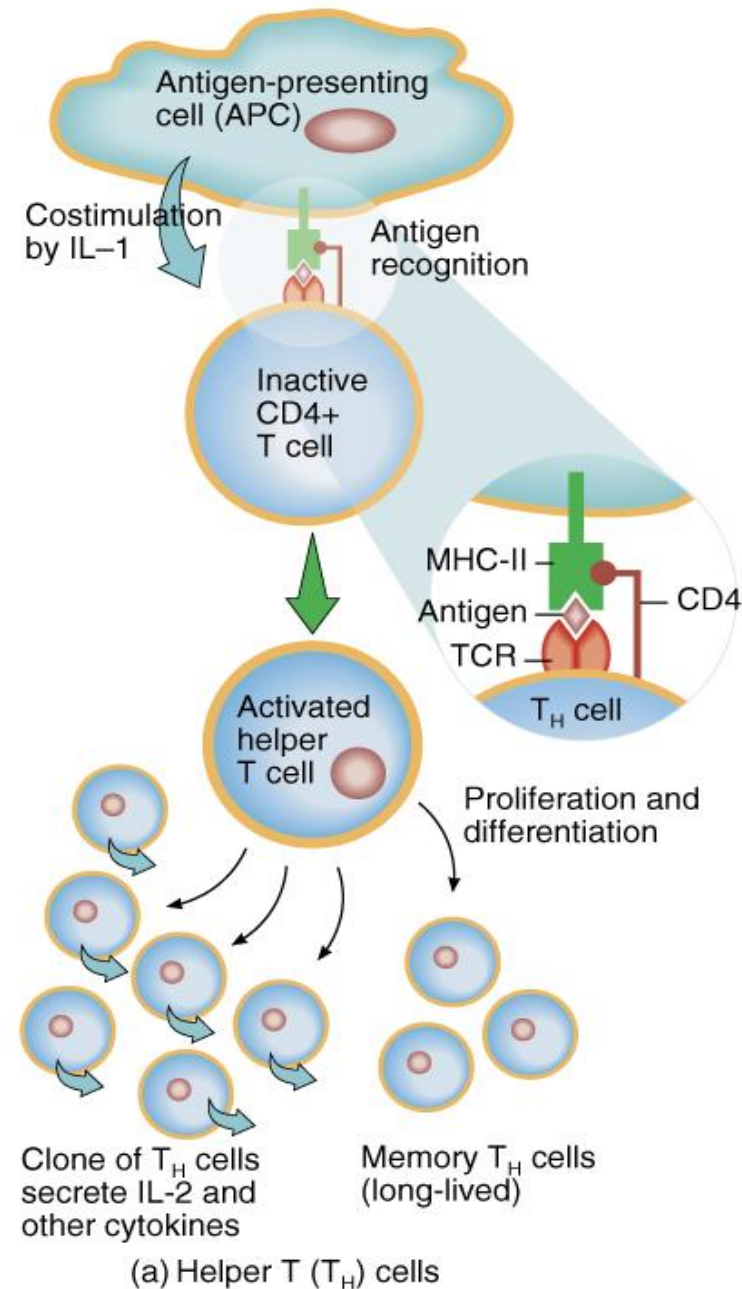
Antigen-Presenting Cells (APCs)



ells

Activation, Proliferation & Differentiation of Helper T Cells

- Receptor on CD4 cell binds to foreign antigen fragment associated with MHC-II
- Costimulation with interleukin
- Proliferates & differentiates into population (clone) of T_H cells and long-lived memory T_H cells



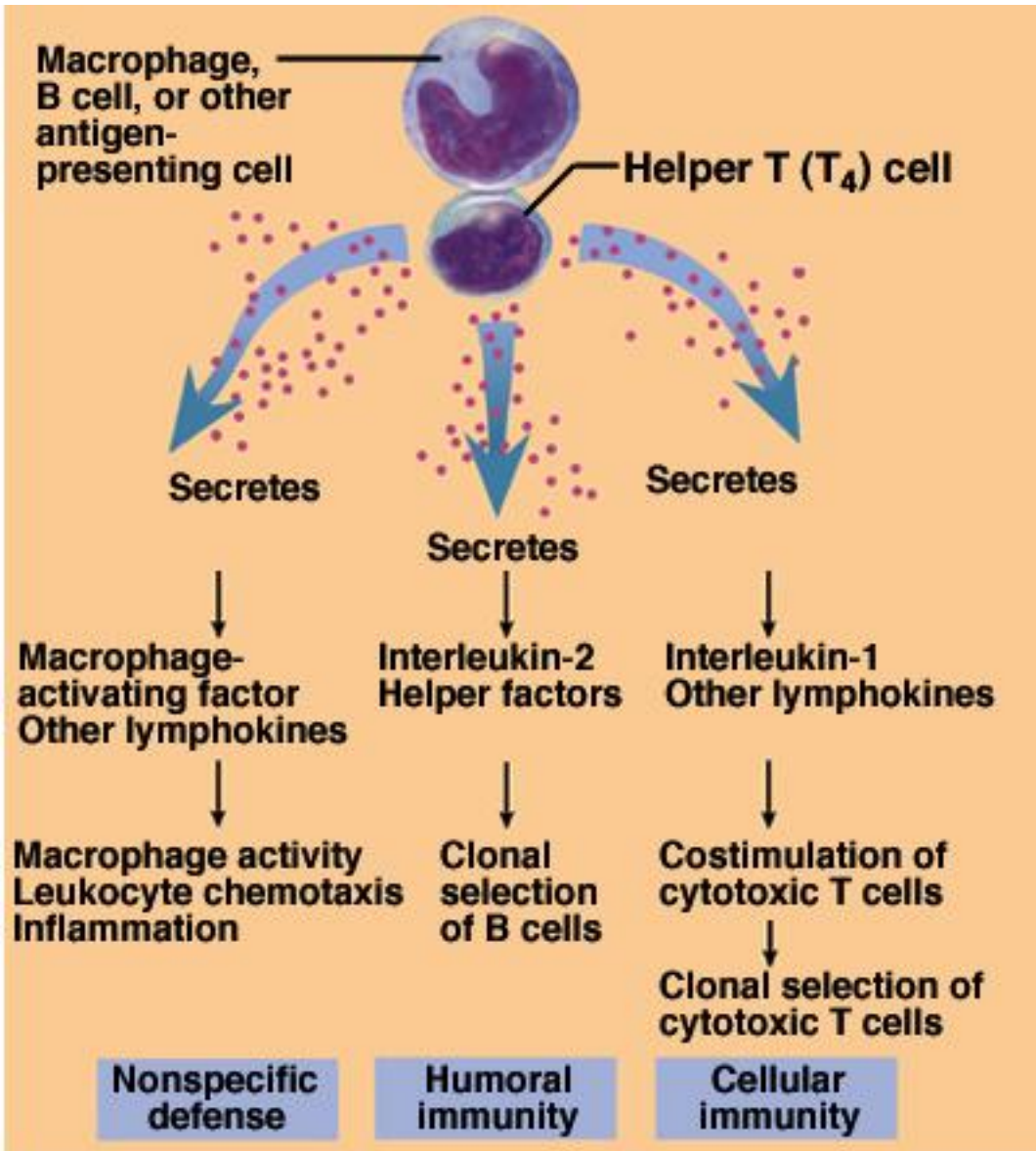
T_H cell Recognition

- Antigen presentation and T cell activation
 - role of **MHC-II** proteins
 - found only on antigen presenting cells
 - stimulate **helper T cells (CD4 cells)**
- T_H cell activation
 - 1.binding of **helper T cells (CD4 cells)** to abnormal peptides **ingested** and displayed on MHC-II and
 - 2.costimulation via a cytokine

Helper T Cells

- Display CD4 on surface so also known as T₄ cells or T_H cells
- Recognize antigen fragments associated with MHC-II molecules & activated by APCs
- Function is to costimulate all other lymphocytes
 - secrete cytokines (interleukin-2)
 - autocrine function in that it costimulates itself to proliferate and secrete more interleukin (positive feedback effect causes formation of many more helper T cells)

Attack Phase: Role of Helper T Cells



- Helper T cells
 - secretes interleukins
 - coordinate humoral and cellular immunity

Memory T Cells

- T cells from a clone that did not turn into cytotoxic T cells during a cell-mediated response
- T cell recall response
 - Available for swift response if a 2nd exposure should occur

Suppressor T Cells

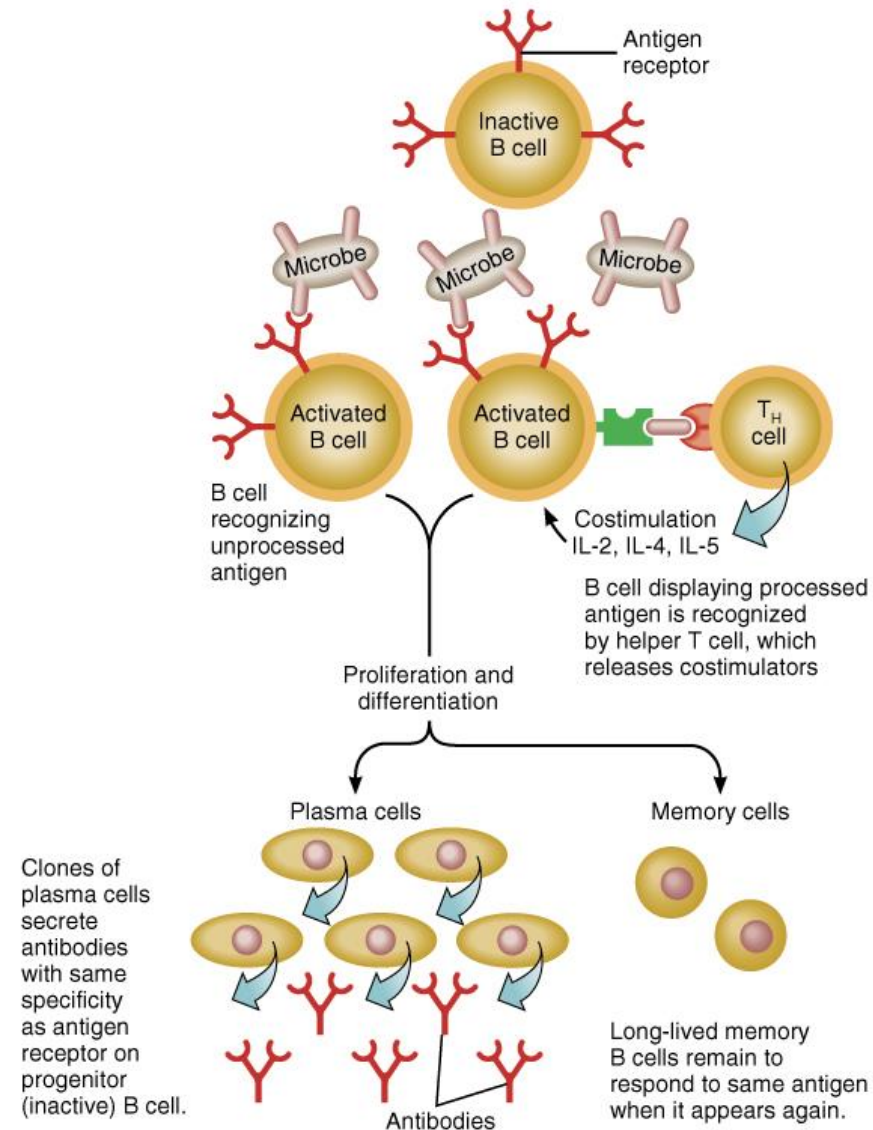
- As pathogen disappears, slows down immune reaction
- Prevents autoimmune diseases

Antibody-Mediated Immunity

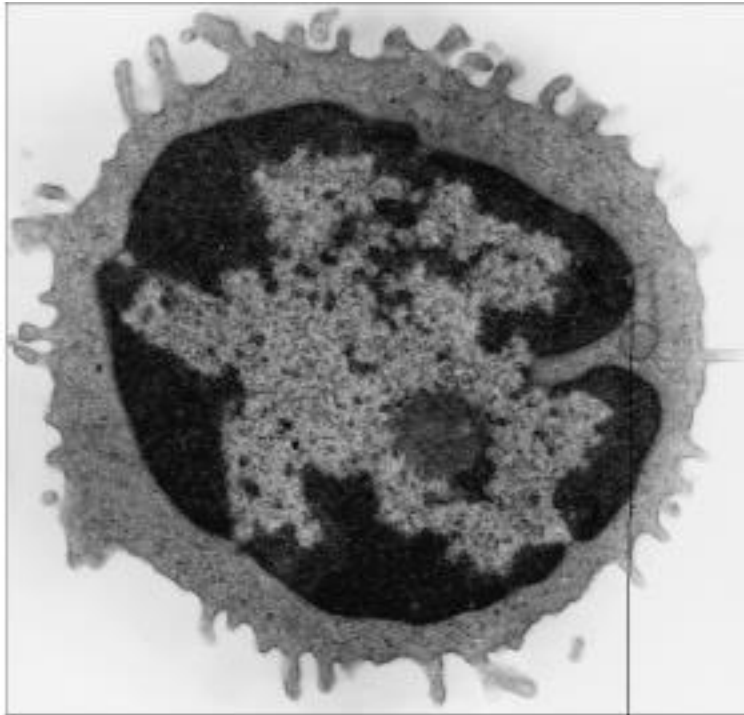
- Millions of different B cells that can recognize different antigens and respond
- B cells sit still and let antigens be brought to them
 - stay put in lymph nodes, spleen or peyer's patches
- Once activated, differentiate into plasma cells that secrete antibodies
- Antibodies circulate in lymph and blood
 - combines with epitope on antigen similarly to key fits a specific lock

Activation, Proliferation, & Differentiation of B Cells

- B cell receptors bind to antigen
-- response more intense if on APC
- Helper T cell costimulates
- Rapid cell division & differentiation occurs
 - long-lived memory cells
 - clone of plasma cells
 - produce antibody at 2000 molecules/sec for 4-5 days
 - secrete only one kind antibody
- Antibody enters the circulation to attack antigen

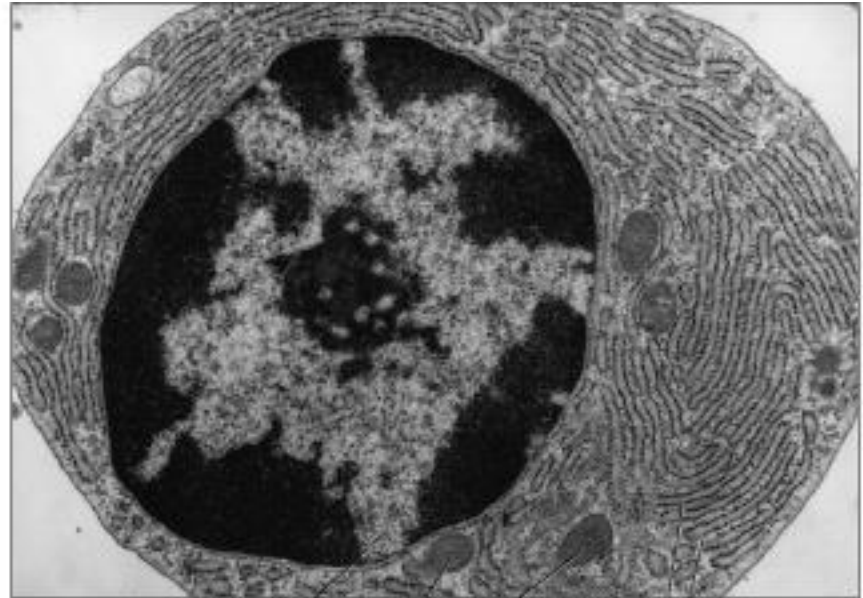


B cell



Nucleus

Plasma cell



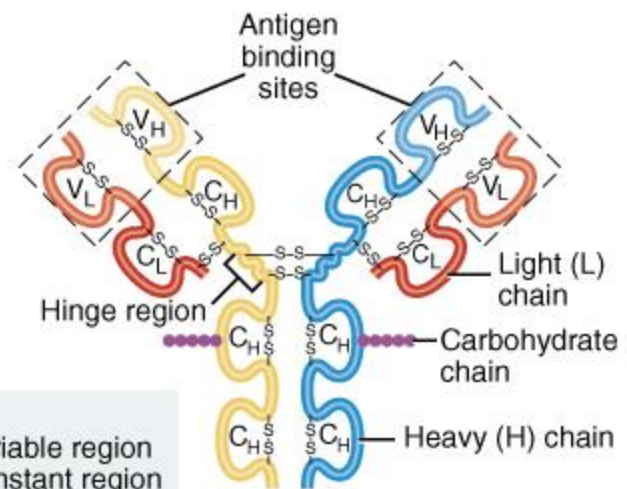
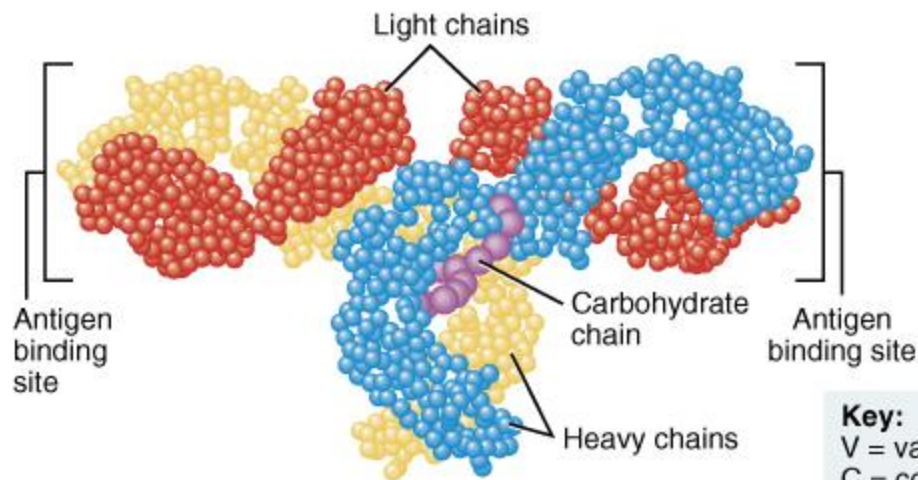
Nucleus

Mitochondria

Rough
endoplasmic
reticulum

Antibody Structure

- Glycoproteins called immunoglobulins
 - 4 polypeptide chains -- 2 heavy & 2 light chains
 - hinged midregion lets assume T or Y shape
 - tips are variable regions -- rest is constant region
 - 5 different classes based on constant region
 - IgG, IgA, IgM, IgD and IgE
 - tips form antigen binding sites



Antibody Classes

- By amino acid sequences of C region of antibody
- IgA: monomer in plasma; dimer in mucus, saliva, tears, milk, intestinal secretions, prevents adherence to epithelia
- IgD: monomer; B cell membrane antigen receptor
- IgE: monomer; tonsils, skin, mucous membranes; stimulates release of histamines, attracts eosinophils
- IgG: monomer; 75-85% circulating, crosses placenta to fetus, secondary immune response, binds complement
- IgM: monomer; B cell membrane, antigen receptor; pentamer in plasma, 1^o immune response, agglutination

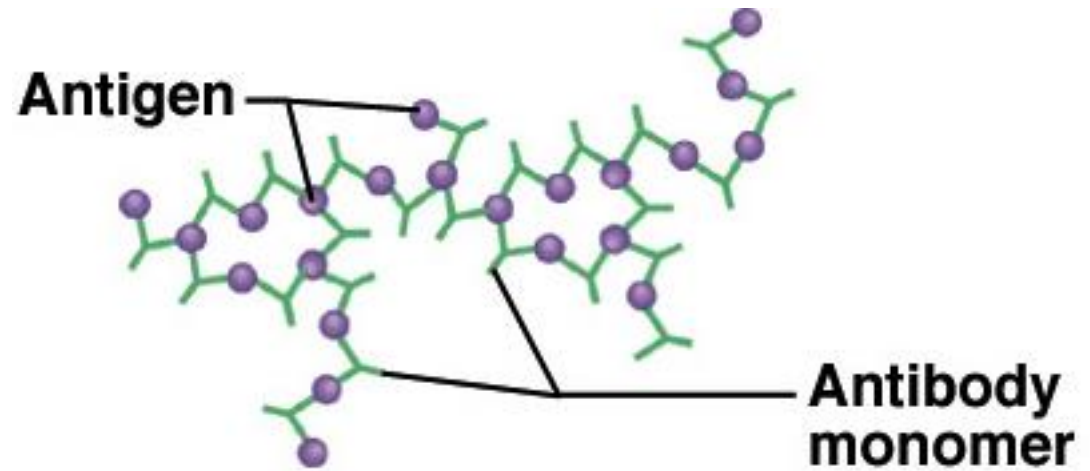
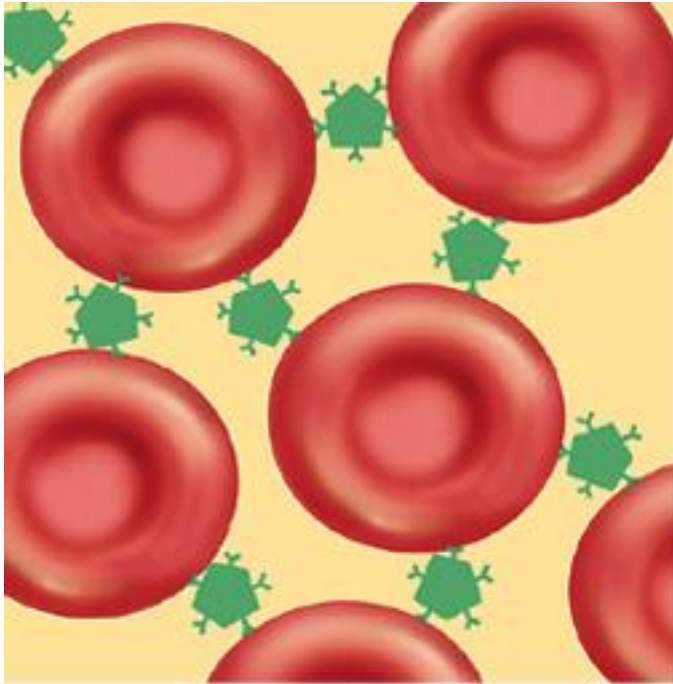
Antibody Diversity

- Immune system produces as many as 2 million different antibodies
- Somatic recombination
 - DNA segments shuffled and form new combinations of base sequences to produce antibody genes

Antibody Actions

- Neutralization of antigen by blocking effects of toxins or preventing its attachment to body cells
- Immobilize bacteria by attacking cilia/flagella
- Agglutinate & precipitate antigens by cross-linking them causing clumping & precipitation
- Complement activation
- Enhancing phagocytosis through precipitation, complement activation or opsonization (coating with special substance)

Agglutination and Precipitation

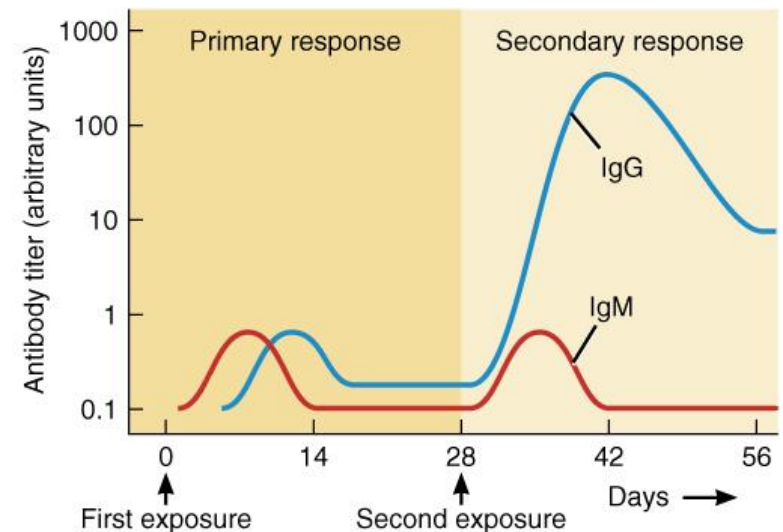


Monoclonal Antibodies

- Antibodies against a particular antigen can be harvested from blood
 - different antibodies will exist for the different epitopes on that antigen
- Growing a clone of plasma cells to produce identical antibodies difficult
 - fused B cells with tumor cells that will grow in culture producing a hybridoma
 - antibodies produced called monoclonal antibodies
- Used clinically for diagnosis -- strep throat, pregnancy, allergies, hepatitis, rabies, cancer

Immunological Memory

- Primary immune response
 - first exposure to antigen
 - response is steady, slow
 - memory cells may remain for decades
- Secondary immune response with 2nd exposure
 - 1000's of memory cells proliferate & differentiate into plasma cells & cytotoxic T cells
 - antibody titer is measure of memory (amount serum antibody)
 - recognition & removal occurs so quickly not even sick



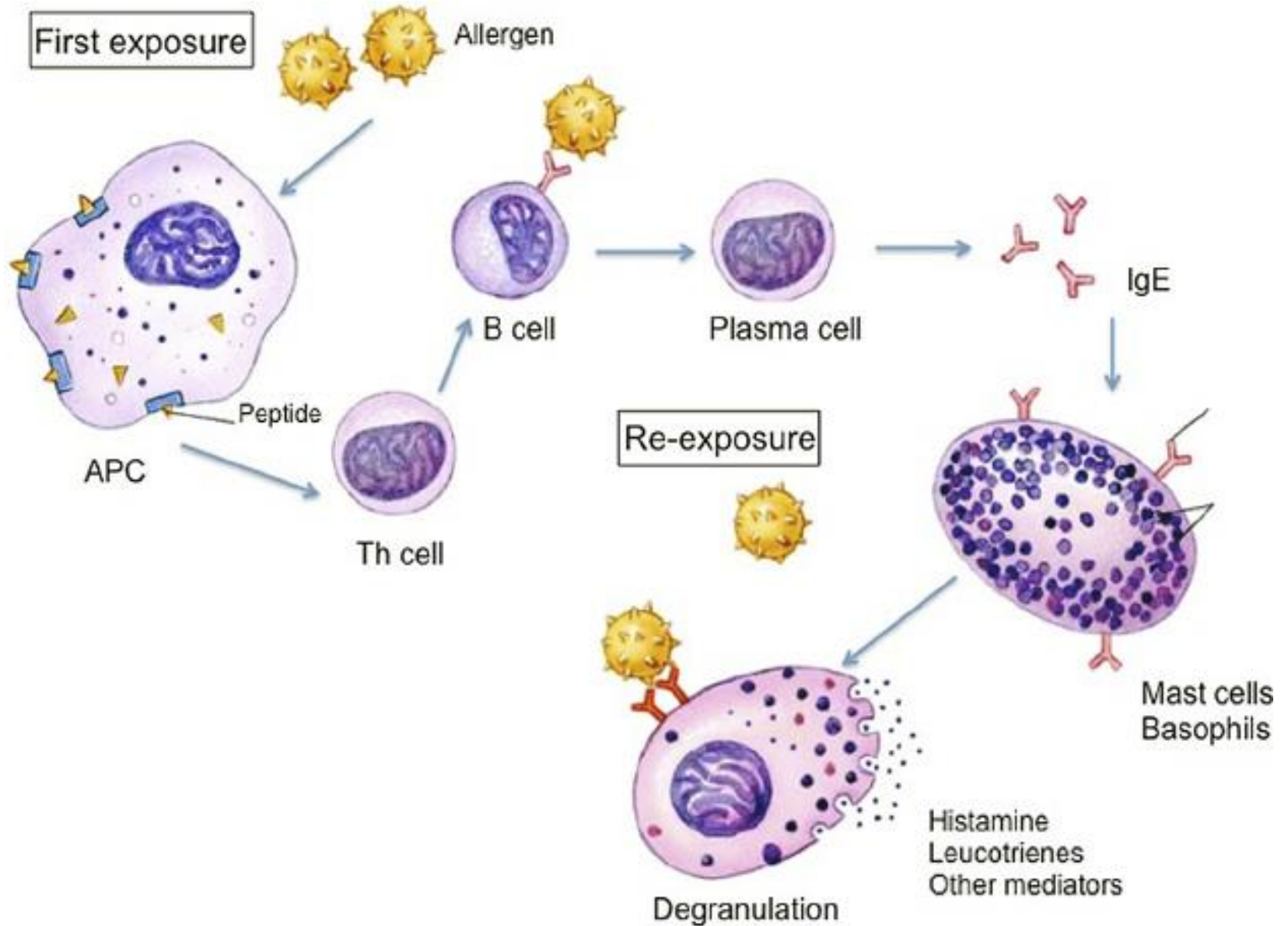
Passive and Active Immunity

- Natural active immunity (produces memory cells)
 - production of one's own antibodies or T cells as a result of infection or natural exposure to antigen
- Artificial active immunity (produces memory cells)
 - production of one's own antibodies or T cells as a result of vaccination
- Natural passive immunity (through placenta, milk)
 - temporary, fetus acquires antibodies from mother
- Artificial passive immunity (snakebite, rabies, tetanus)
 - temporary, injection of immune serum (antibodies)

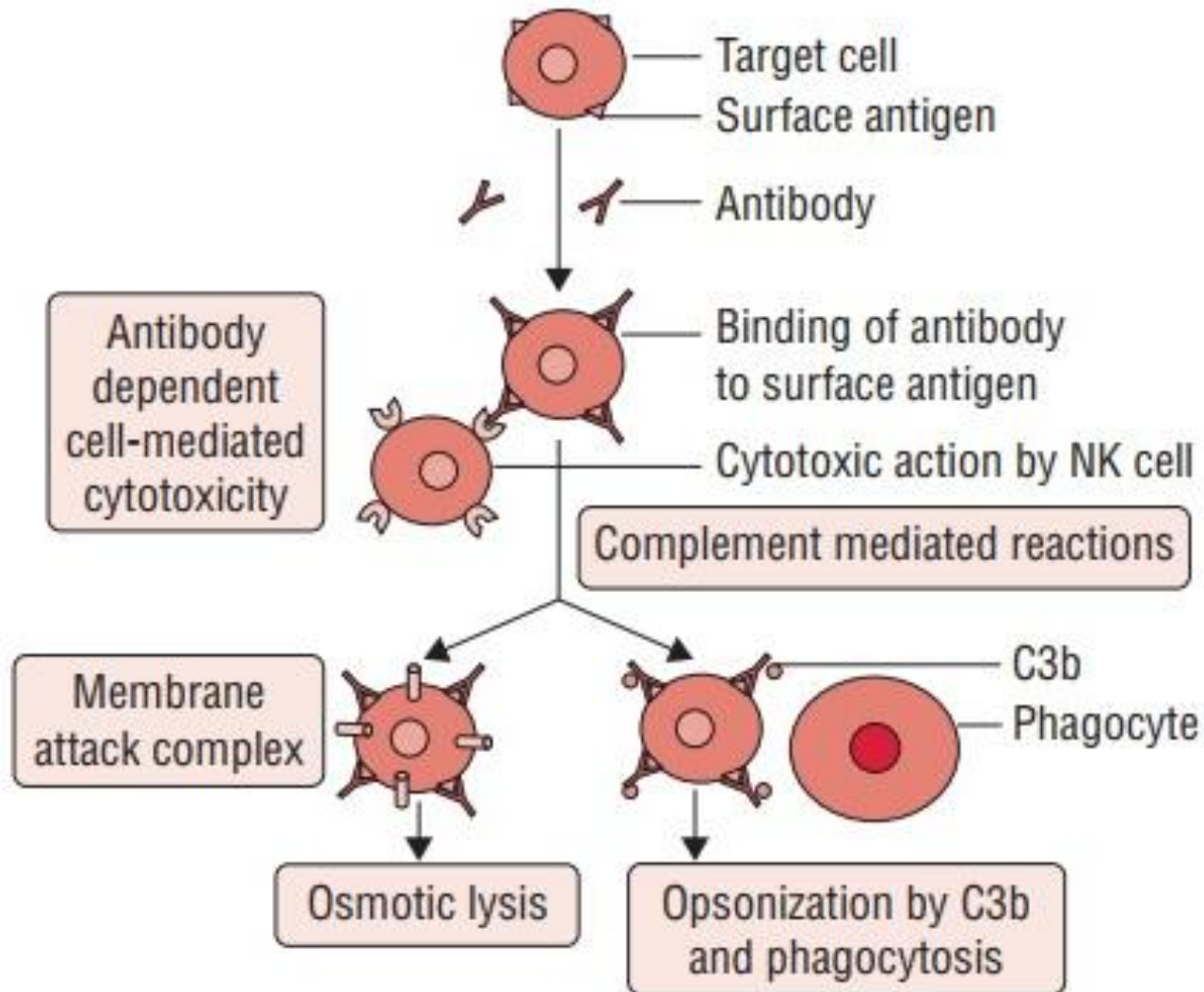
Hypersensitivity (Allergy)

- Excessive immune reaction against antigens that most people tolerate - allergens
- Type I (acute) hypersensitivity (most common)
 - anaphylaxis: occurs in sensitized people, allergen caps IgE on mast cells, basophils; release inflammatory chemicals, cause local edema, mucus hypersecretion, congestion; hives, watery eyes, runny nose are typical
- Asthma (most common chronic illness in children)
 - inhaled allergens, histamines, bronchiole constriction
- Anaphylactic shock: bronchiolar constriction, dyspnea, vasodilation, shock, death; treatment- epinephrine

Type I Hypersensitivity



Type II Hypersensitivity

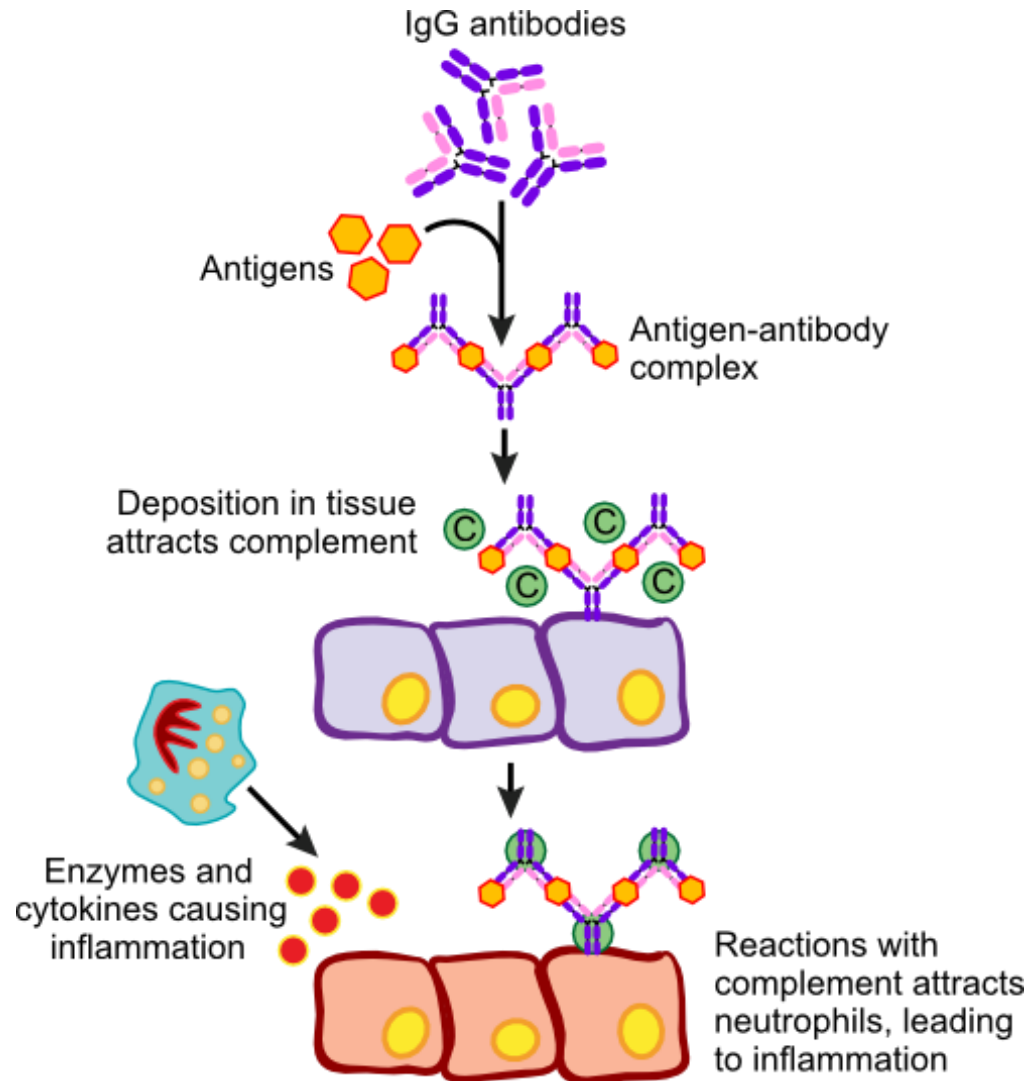


Type II Hypersensitivity

(Antibody-Dependent Cytotoxic)

- IgG or IgM binds to antigens on cells and lyses them through complement fixation; transfusion reaction
- IgG or IgM binds to cell surface receptors and either interferes with (myasthenia gravis, erythroblastosis fetalis) or overstimulates cell

Type III Hypersensitivity

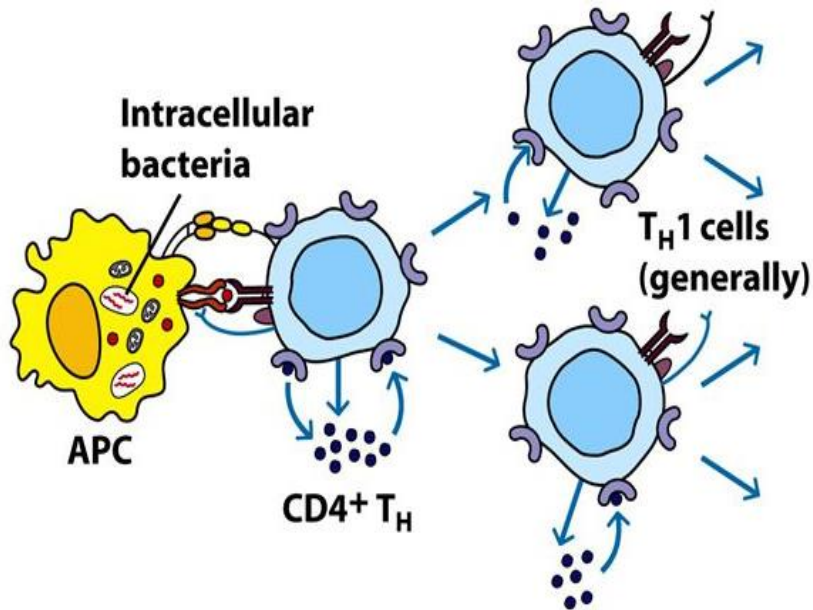


Type III Hypersensitivity (Immune Complex)

- Widespread antigen-antibody complexing
- Complexes trigger intense inflammation, involved in acute glomerulonephritis and in systemic lupus erythematosus

Type IV Hypersensitivity

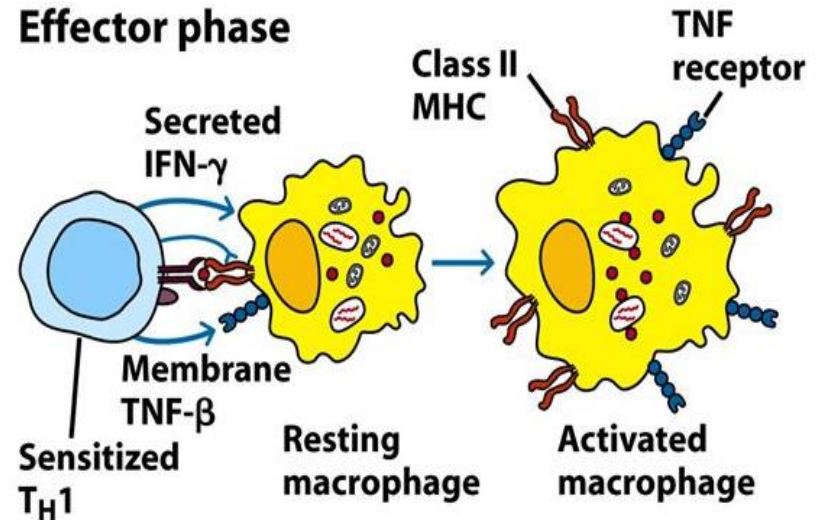
Sensitization phase



Antigen-presenting cells: Macrophages
Langerhans cells

DTH-mediating cells:
T_H1 cells generally
CD8 cells occasionally

Effector phase



T_H1 secretions:

Cytokines: IFN-γ, TNF-β,
IL-2,
IL-3, GM-CSF, MIF
Chemokines: IL-8/CXCL8,
MCP-1/CCL2

Effects of macrophage activation:

↑ Class II MHC molecules
↑ TNF receptors
↑ Oxygen radicals
↑ Nitric oxide

Type IV Hypersensitivity (Delayed)

- 12 to 72 hour delay
 - APC's in lymph nodes display antigens to helper T cells, which secrete interferon and lymphokines that activate cytotoxic T cells and macrophages
- Cosmetic and poison ivy allergies - haptens
- TB skin test

Autoimmune Diseases

- Failure of self tolerance
 - cross-reactivity
 - abnormal exposure of self-antigens
 - changes in structure of self-antigens
- Production of autoantibodies

Immunodeficiency Diseases

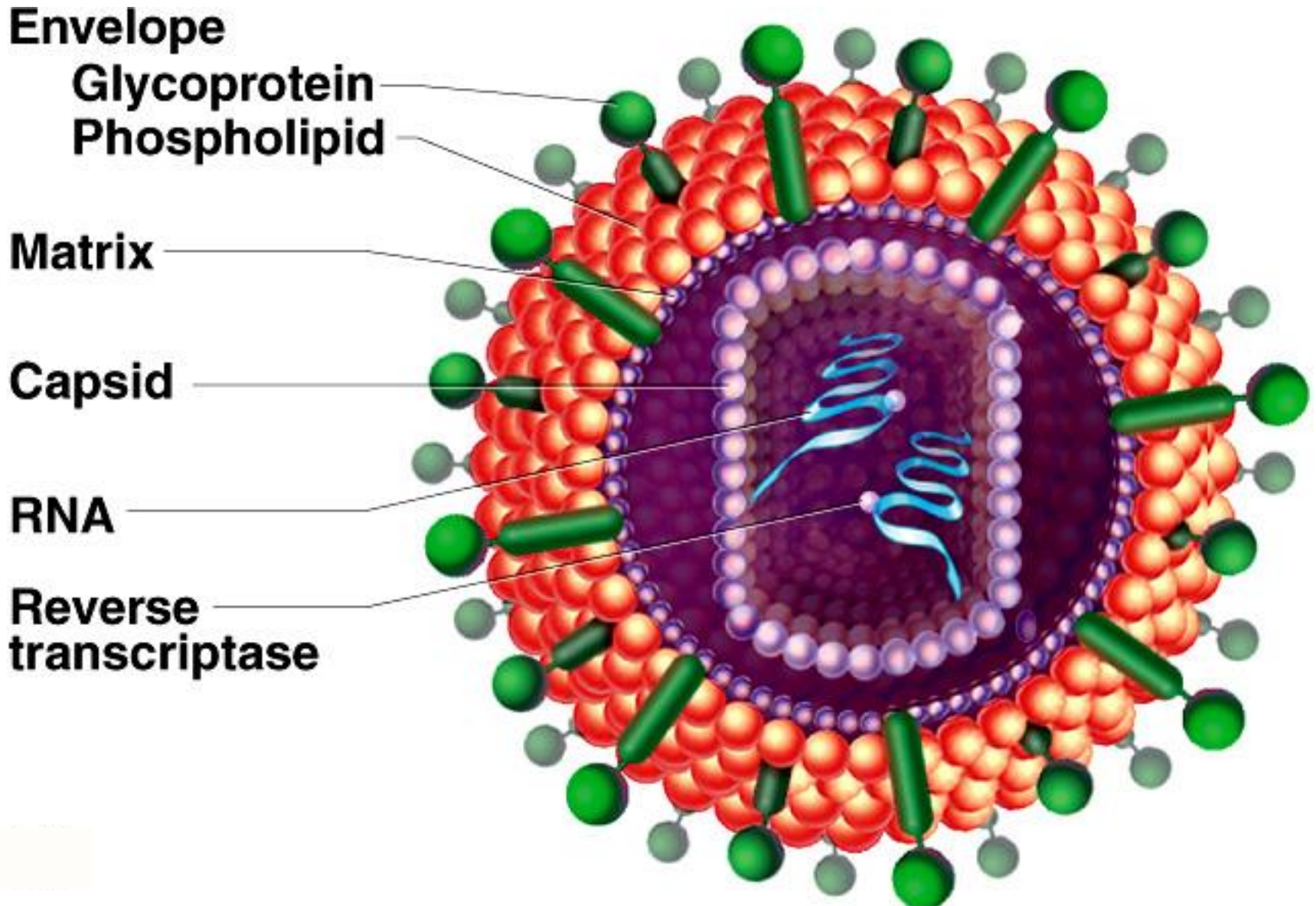
- Severe Combined Immunodeficiency Disease
 - hereditary lack of T and B cells
 - vulnerability to opportunistic infection



Immunodeficiency Diseases

- AIDS
 - HIV structure (*next slide*)
 - invades helper T cells, macrophages and dendritic cells by “tricking” them to internalize viruses by receptor mediated endocytosis
 - **reverse transcriptase (retrovirus)**, uses viral RNA as template to synthesize DNA, new DNA inserted into host cell DNA, may be dormant for months to years

HIV Structure



AIDS

- Signs and symptoms
 - early symptoms: flulike chills and fever
 - progresses to night sweats, fatigue, headache, extreme weight loss, lymphadenitis
 - normal T_H count is 600 to 1,200 cells/ μ L of blood but in AIDS it is < 200 cells/ μ L
 - person susceptible to opportunistic infections (*Toxoplasma*, *Pneumocystitis*, herpes simplex virus, CMV or TB)
 - thrush: white patches on mucous membranes may appear
 - Kaposi sarcoma: cancer originates in endothelial cells of blood vessels causes purple lesions in skin

Kaposi Sarcoma



HIV Transmission

- Through blood, semen, vaginal secretions, breast milk, or across the placenta
- Most common means of transmission
 - sexual intercourse (vaginal, anal, oral)
 - contaminated blood products
 - contaminated needles
- Not transmitted by casual contact
- Undamaged latex condom is an effective barrier to HIV
 - spermicide nonoxynol-9

Tumor Immunotherapy

- Cells with antitumor activity are injected into bloodstream of cancer patient
 - culture patient's inactive cytotoxic T cells with interleukin-2
 - called lymphokine-activated killer cells (LAK)
- Can cause tumor regression, but has severe complications

Aging

- More susceptible to all types of infections and malignancies
- Response to vaccines is decreased
- Produce more autoantibodies
- Reduced immune system function
 - T cells less responsive to antigens
 - age-related atrophy of thymus
 - decreased production of thymic hormones
 - B cells less responsive
 - production of antibodies is slowed